

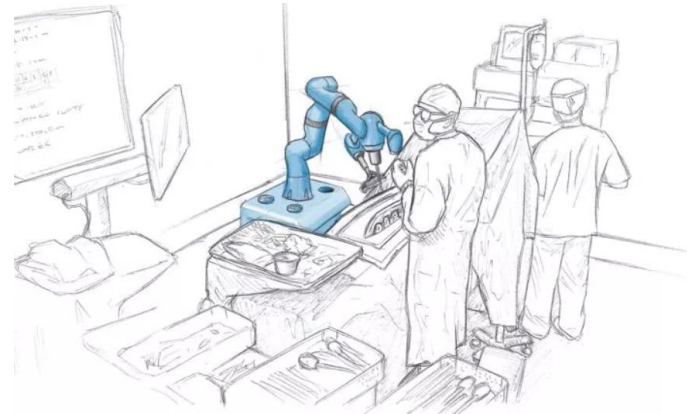
Designing Control Barrier Functions for Safety-critical Systems

Victor Freire

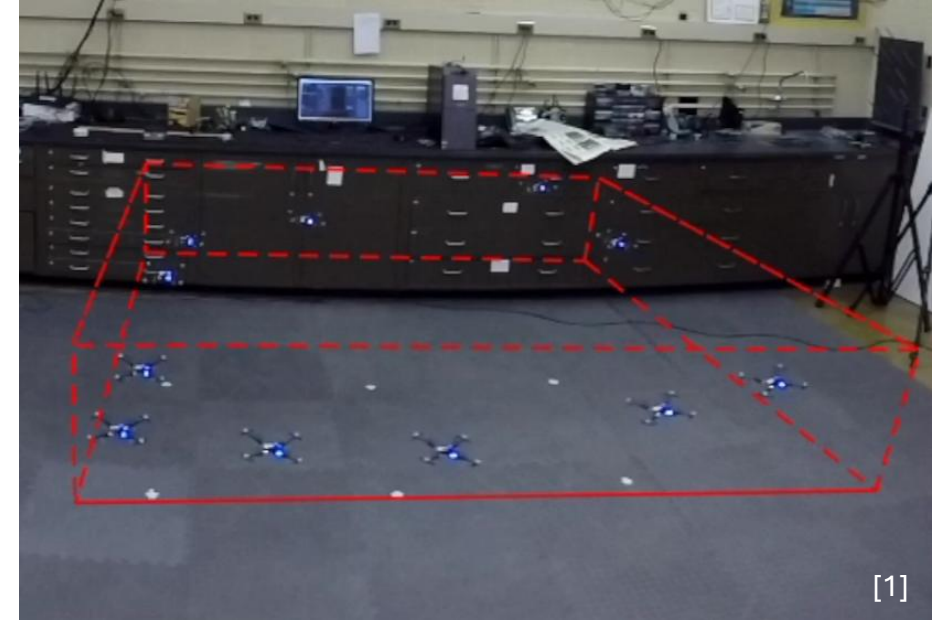
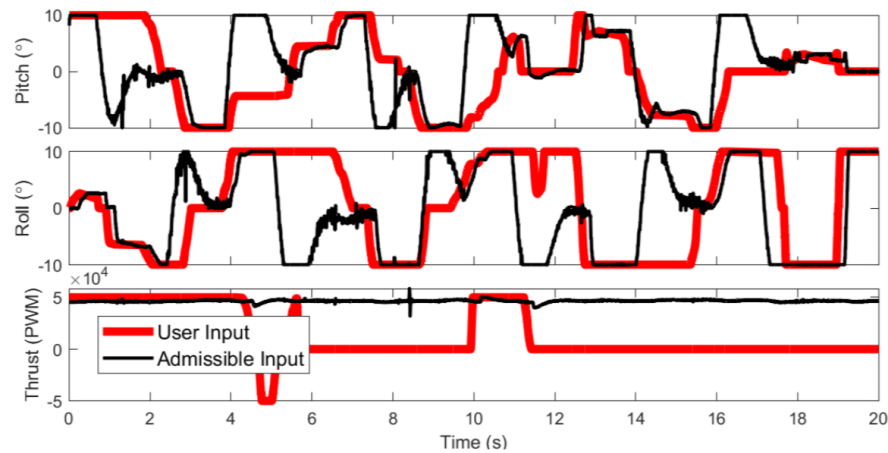
Robotics Seminar

November 5, 2024

Motivation: Safety-critical Applications

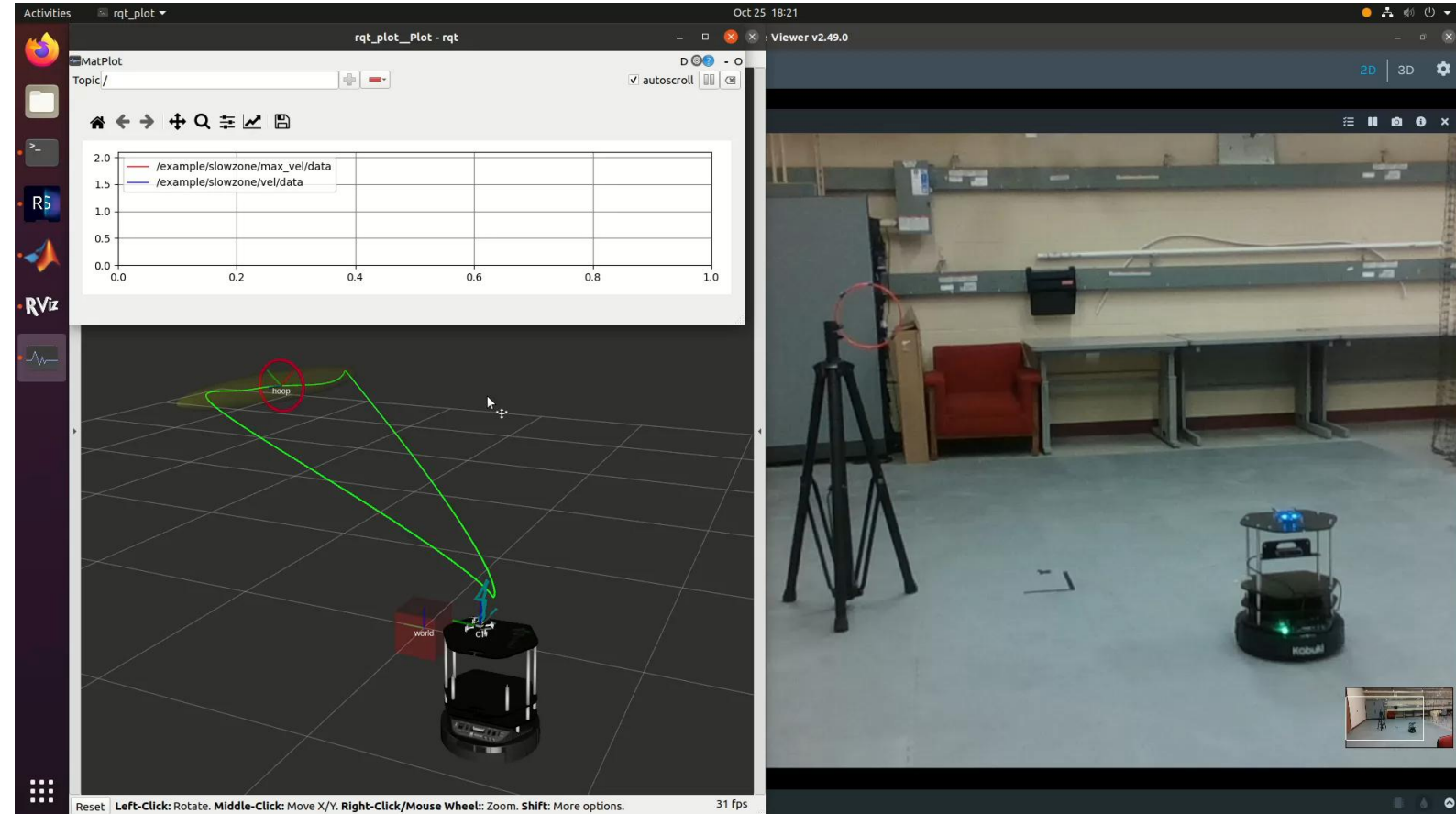


Motivation: Safety-critical Applications



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Example from my master's



[2]

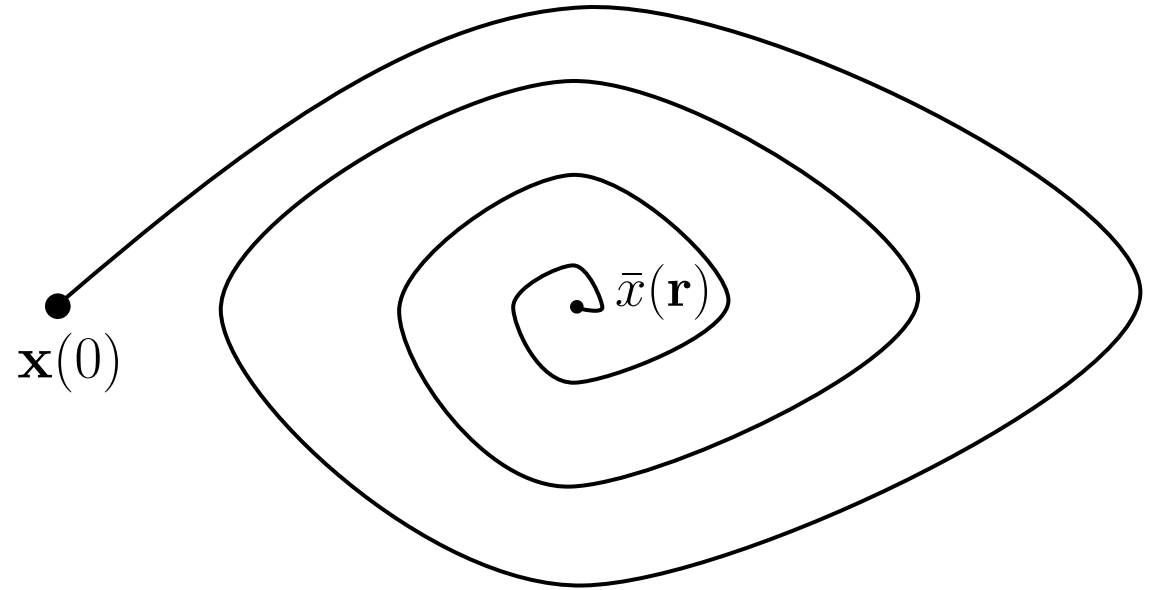
Control Barrier Functions (CBFs)

Continuous-time nonlinear system:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}$$

Nominal controller:

$$\mathbf{u} = \kappa(\mathbf{x}, \mathbf{r})$$



Control Barrier Functions (CBFs)

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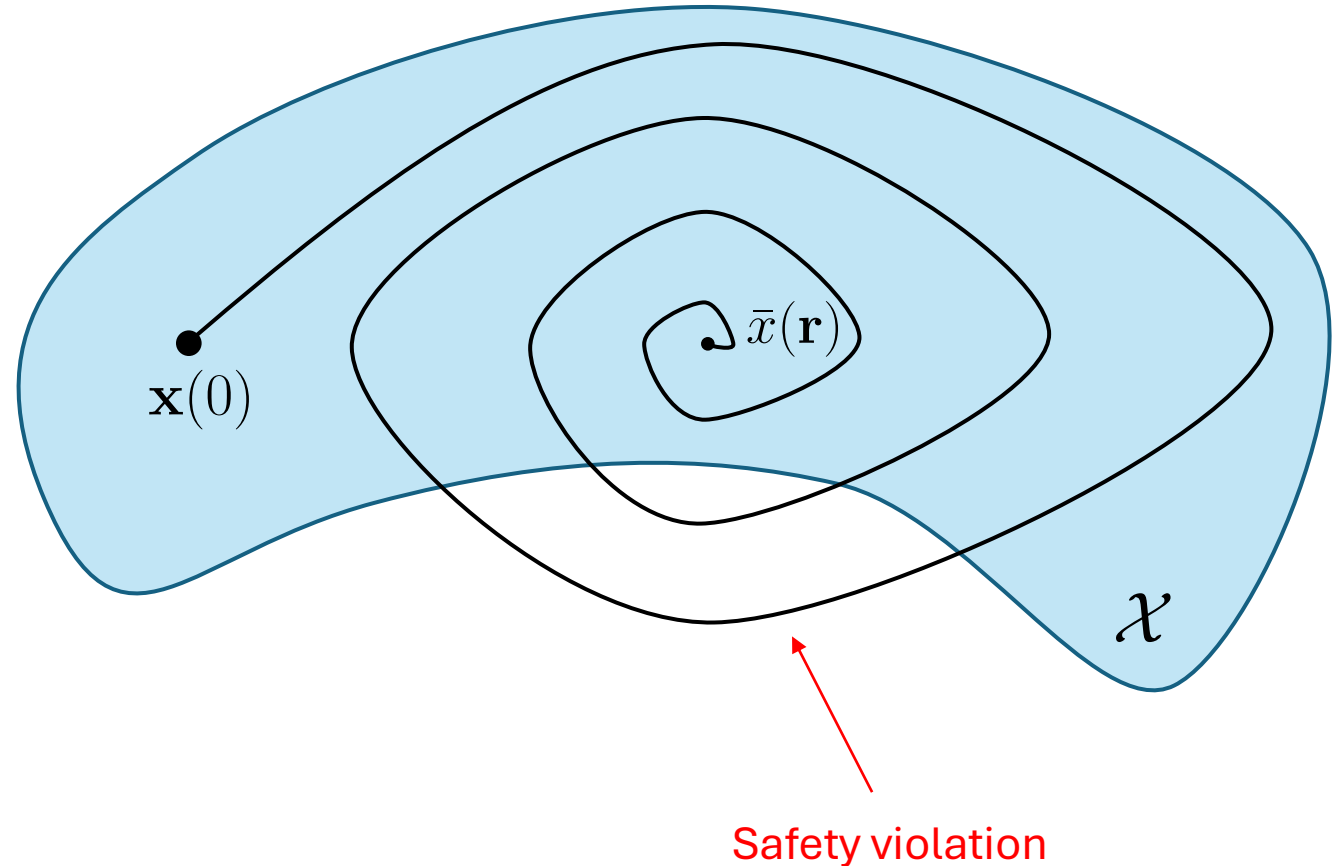
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Safety set $\mathcal{X}$

Safety requirement: $\mathbf{x}(t) \in \mathcal{X}, \forall t$



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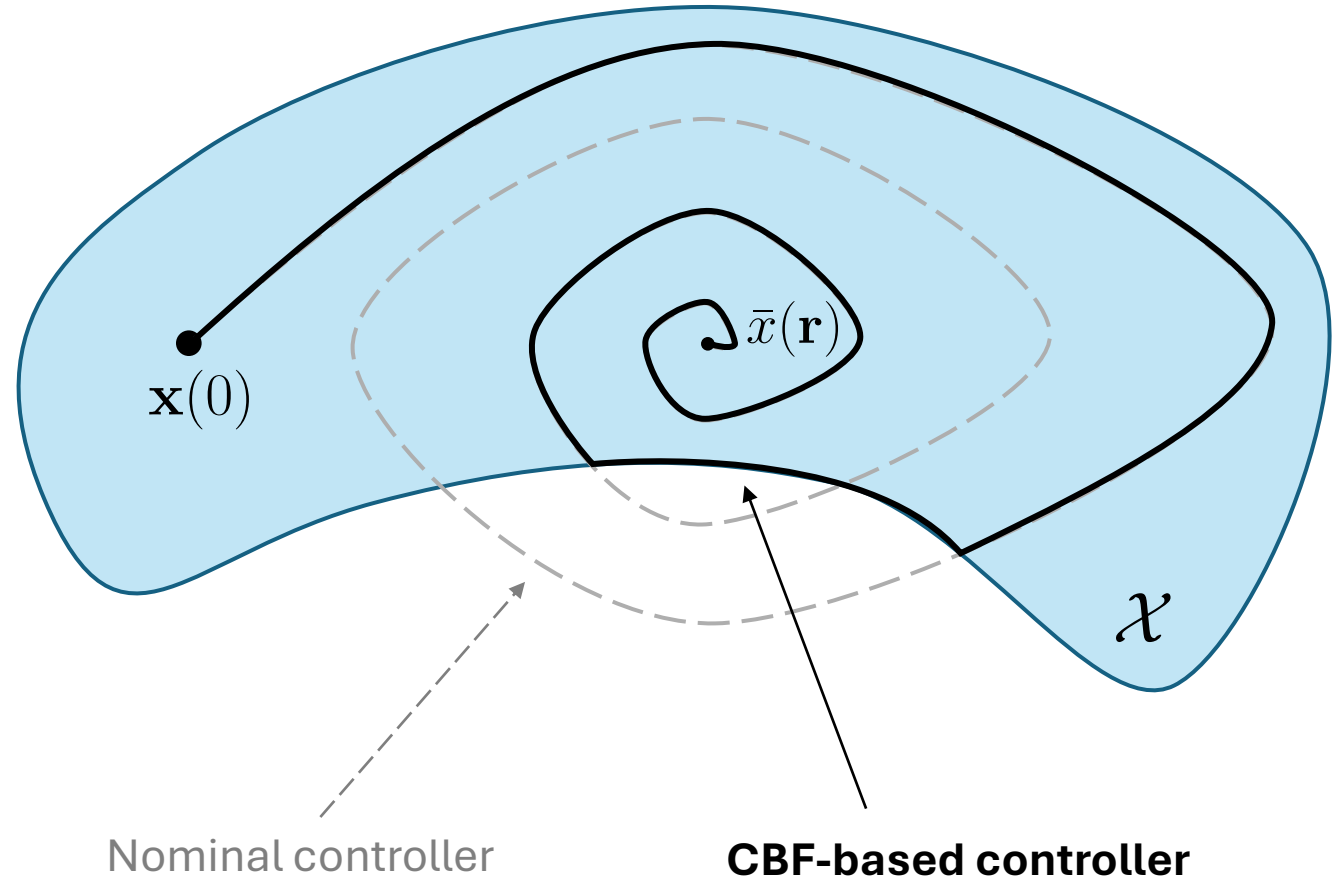
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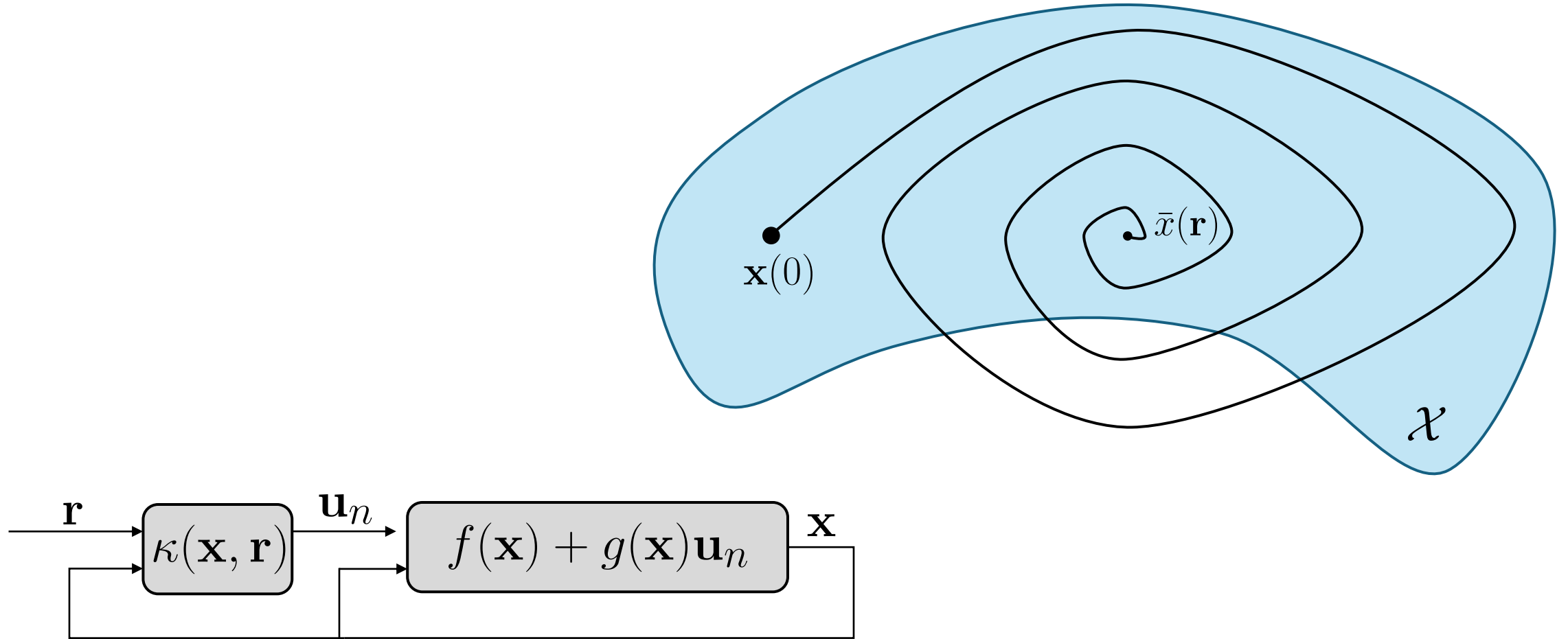
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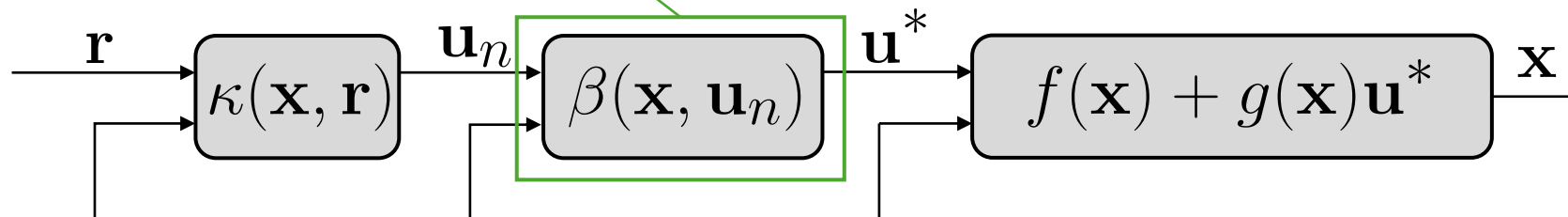
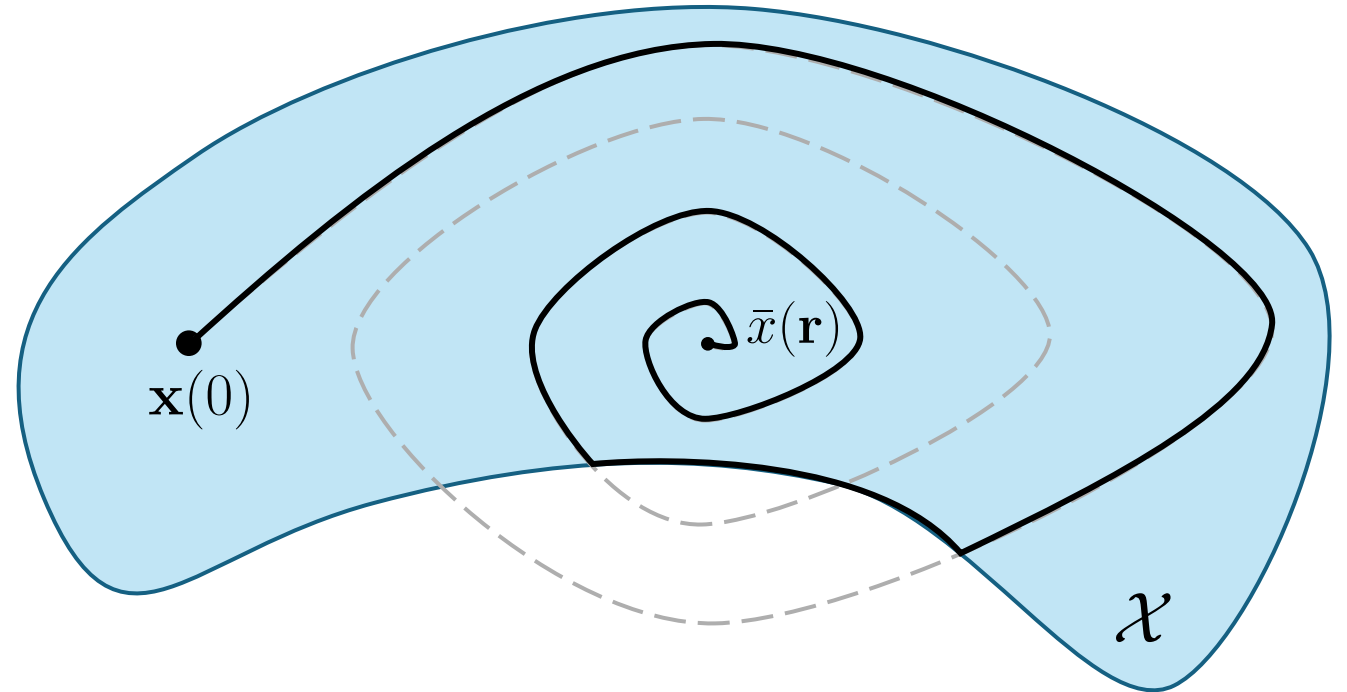
Control Barrier Functions (CBFs)



Control Barrier Functions (CBFs)

$$\mathcal{X} = \{\mathbf{x} \in \mathbb{R}^n \mid h(\mathbf{x}) \geq 0\}$$

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 \\ \text{s.t.} \quad & \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x})) \end{aligned}$$



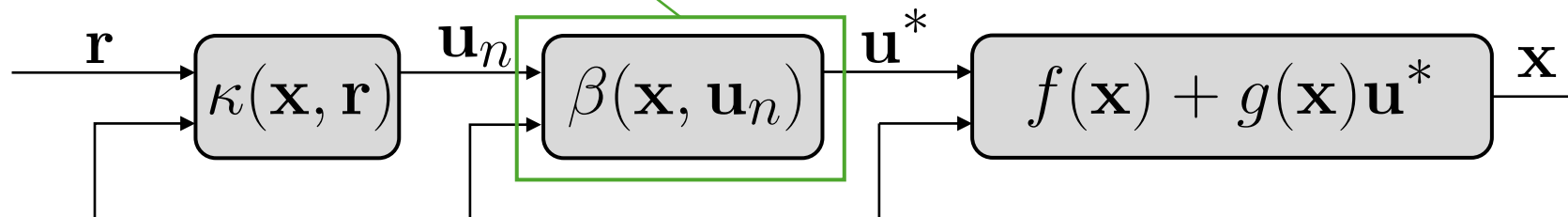
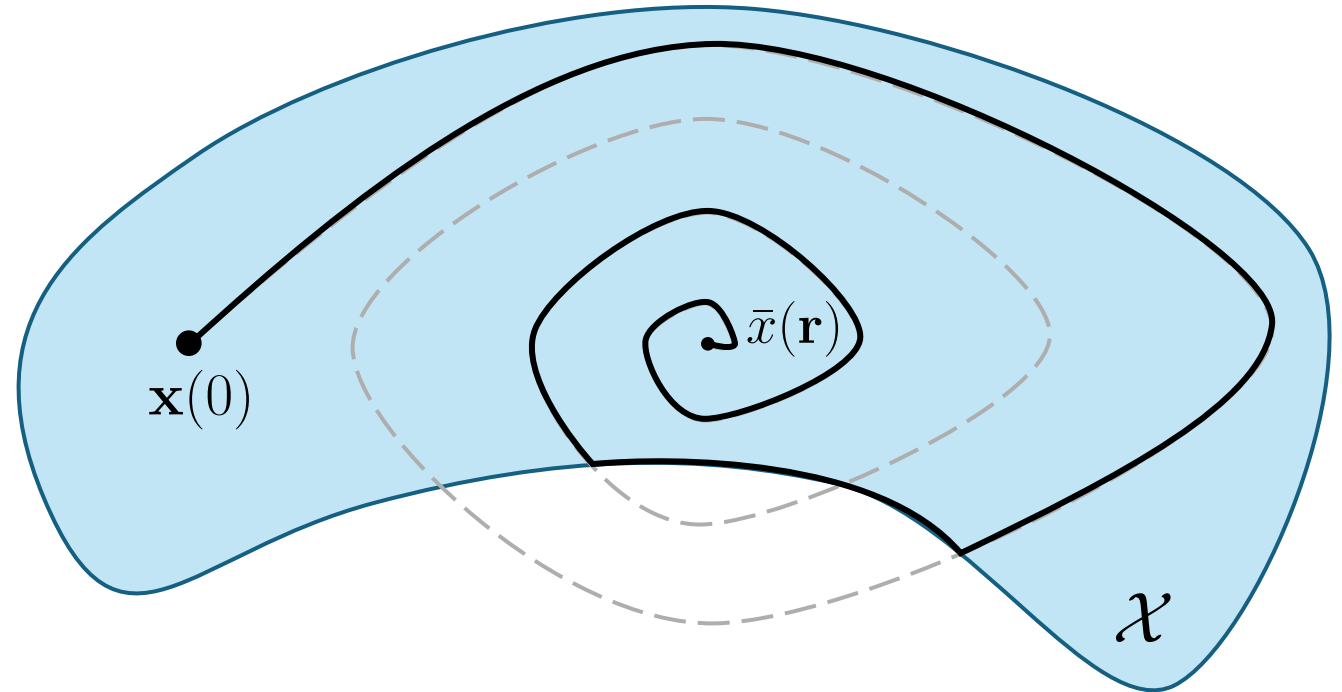
Control Barrier Functions (CBFs)

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$$\dot{h}(\mathbf{x}, \mathbf{v}) = L_f h(\mathbf{x}) + L_g h(\mathbf{x}) \mathbf{u}$$

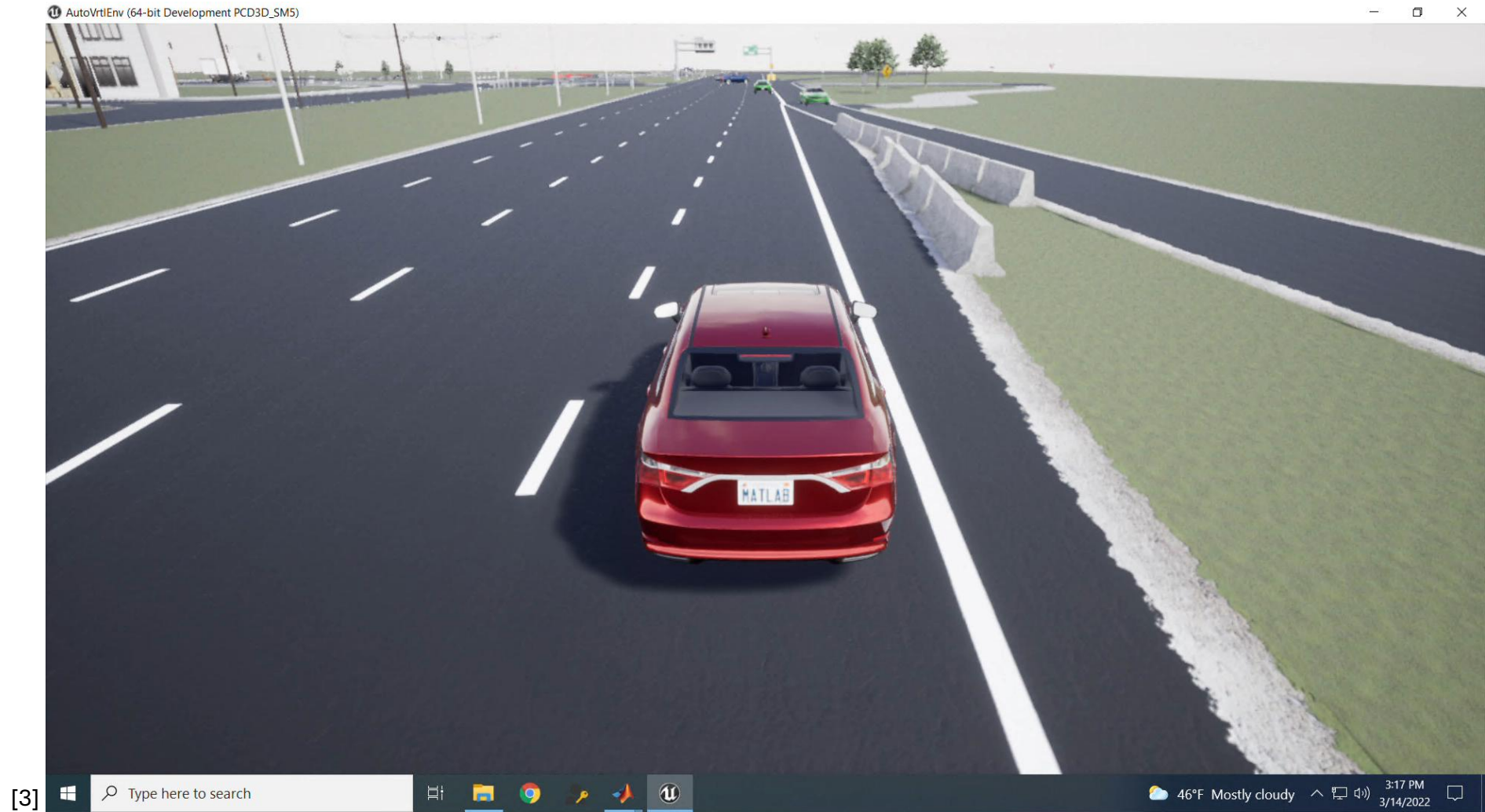
Quadratic Program (QP)

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 \\ \text{s.t.} \quad & \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x})) \end{aligned}$$



CBF Example: Adaptive Cruise Control

Simulation in Mcity
(University of Michigan)



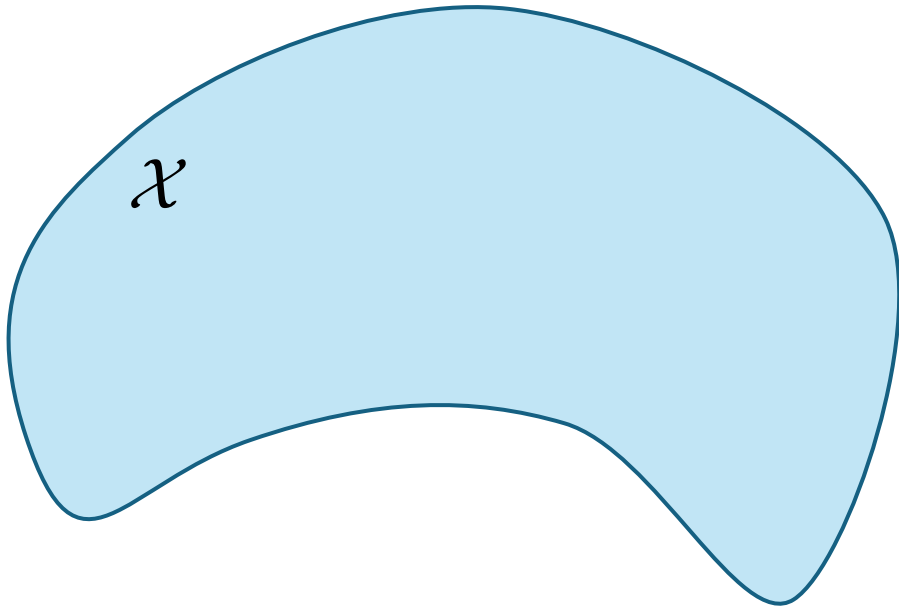
[3]



[3] Freire, Xu, RAL 2022

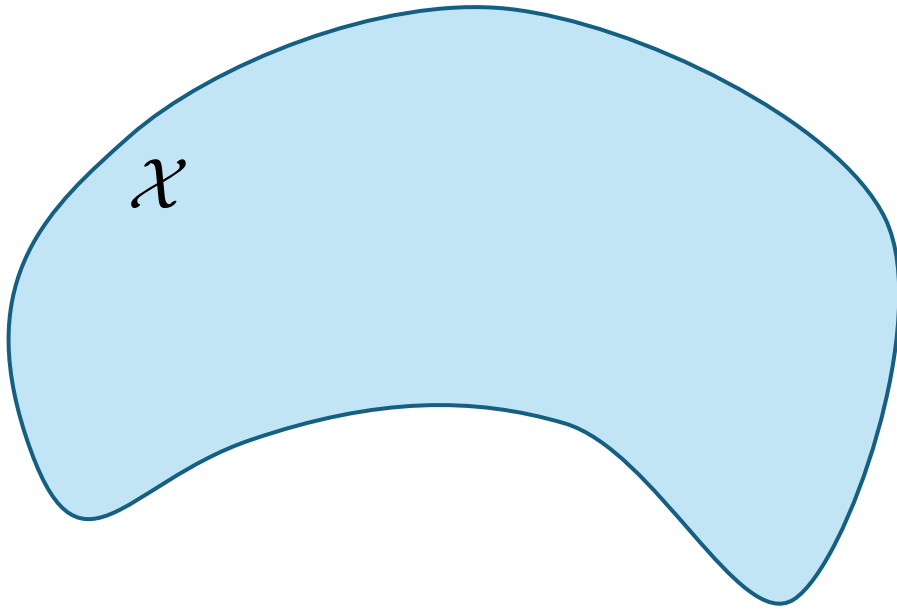
Victor Freire
Slide 11 of 56

Control Barrier Functions (CBFs)



Candidate CBF: $h \in C^1$ such that
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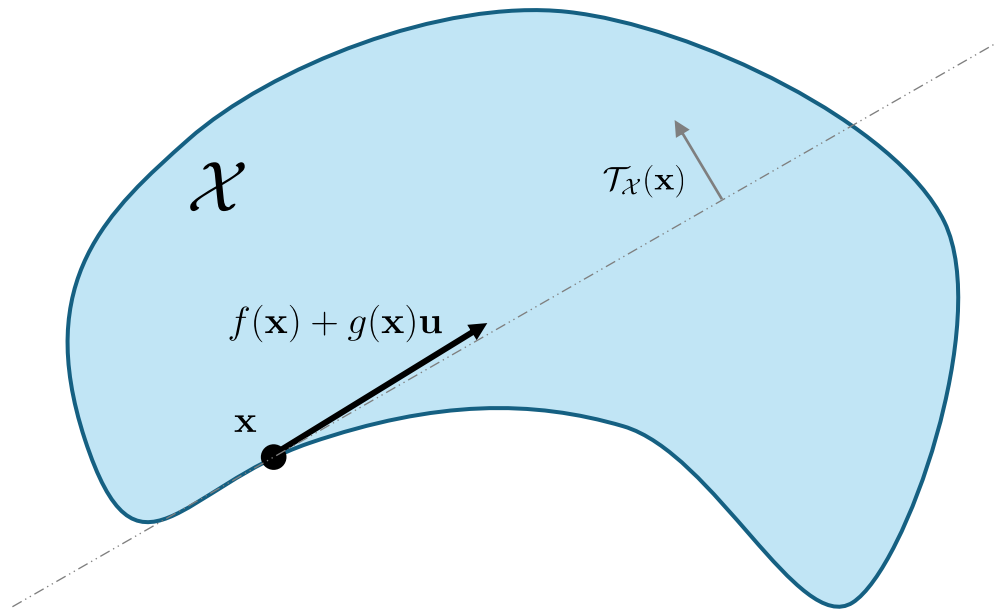
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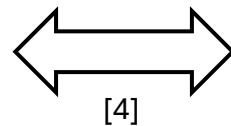
CBF condition: $\exists \alpha \in K_\infty$
$$\sup_{\mathbf{u} \in \mathcal{U}} \left[\dot{h}(\mathbf{x}, \mathbf{u}) \right] \geq -\alpha(h(\mathbf{x})), \forall \mathbf{x} \in \mathcal{X}$$

Control Barrier Functions (CBFs)



Nagumo's control invariance conditions:

$$\forall \mathbf{x} \in \partial \mathbf{X}, \exists \mathbf{u} \in \mathcal{U}, f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \in \mathcal{T}_X(\mathbf{x})$$



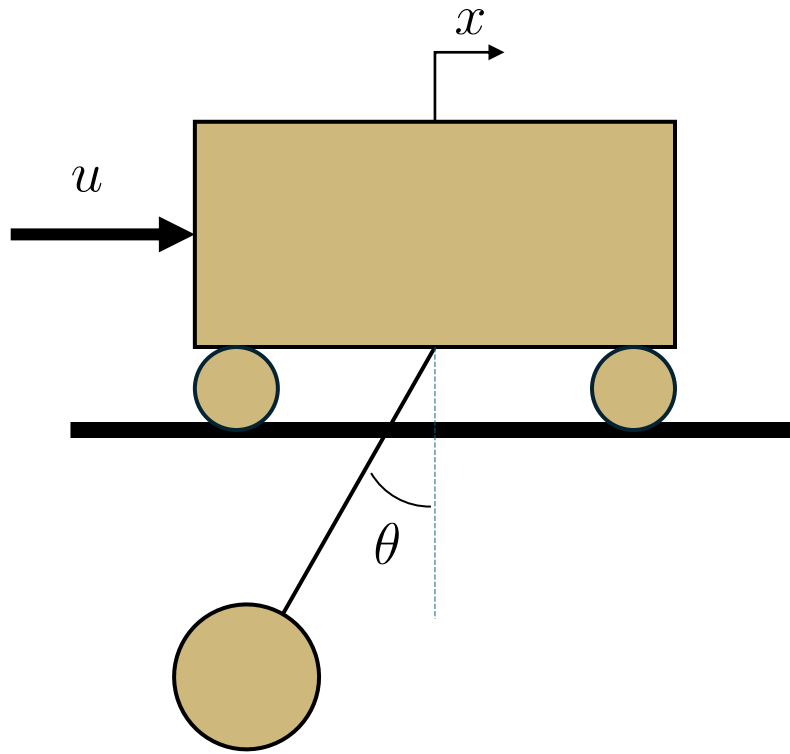
[4]

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Example: Overhead Crane

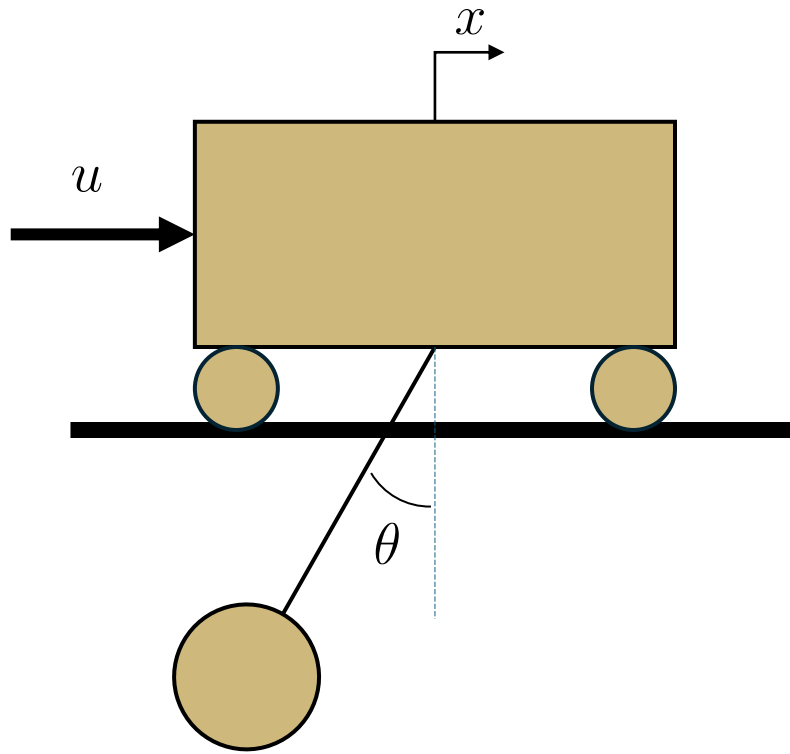


$$\mathbf{q} = [x; \theta]$$

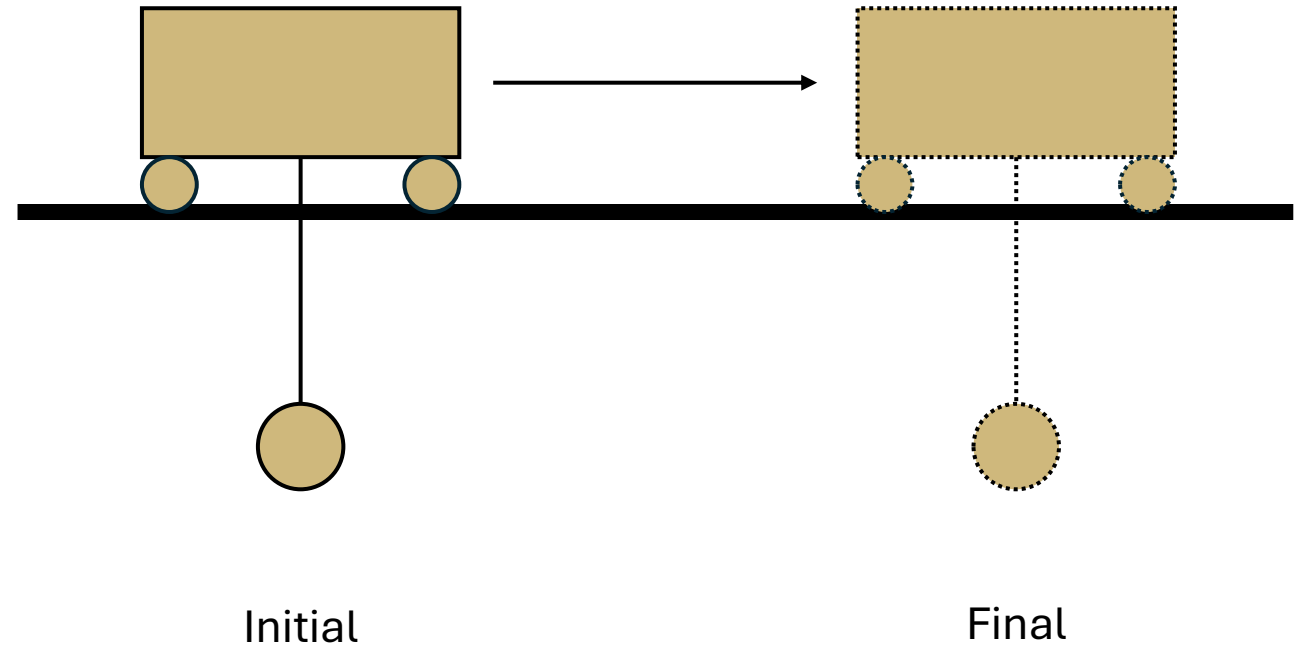
$$M(\mathbf{q})\ddot{\mathbf{q}} + V_m(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + G(\mathbf{q}) = Bu,$$

$$M(\mathbf{q}) = \begin{bmatrix} m_c + m_p & -m_p L \cos \theta \\ -m_p L \cos \theta & m_p L^2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$
$$V_m(\mathbf{q}, \dot{\mathbf{q}}) = \begin{bmatrix} 0 & m_p L \dot{\theta} \sin \theta \\ 0 & 0 \end{bmatrix}, \quad G(\mathbf{q}) = \begin{bmatrix} 0 \\ m_p g L \sin \theta \end{bmatrix},$$

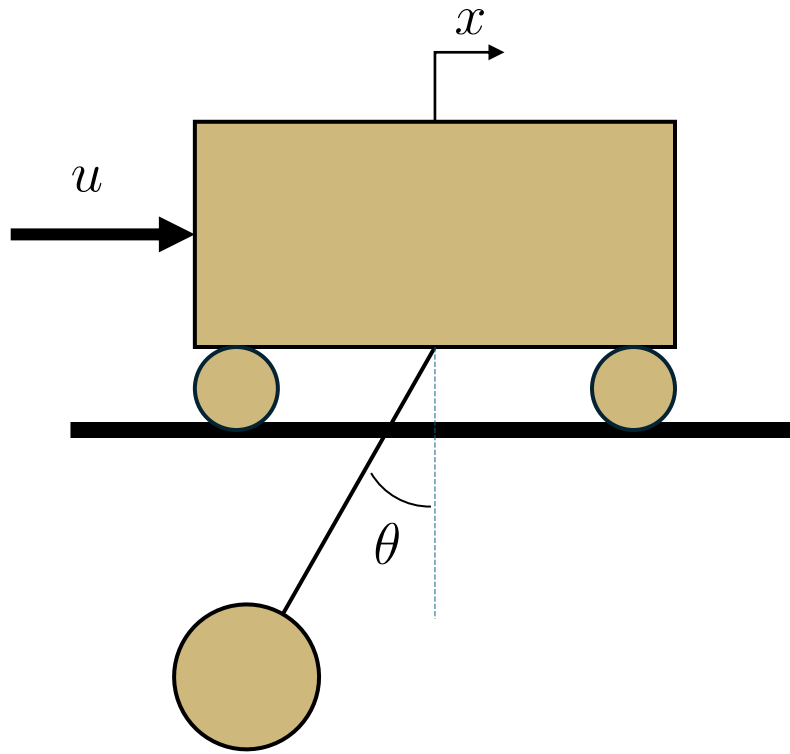
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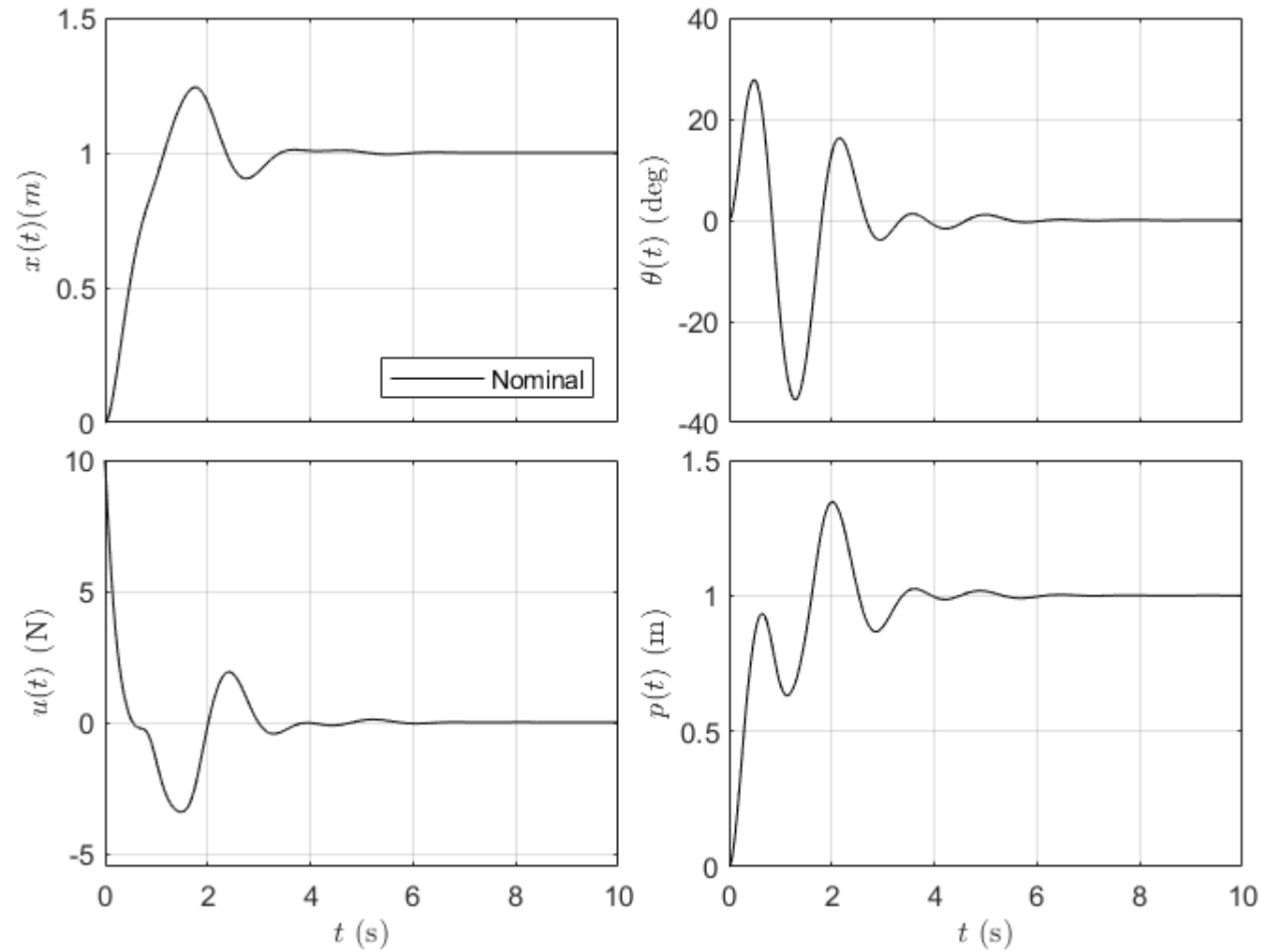
PD Control law $\kappa(\mathbf{x}, r) = -k_p(x - r) - k_d\dot{x}$



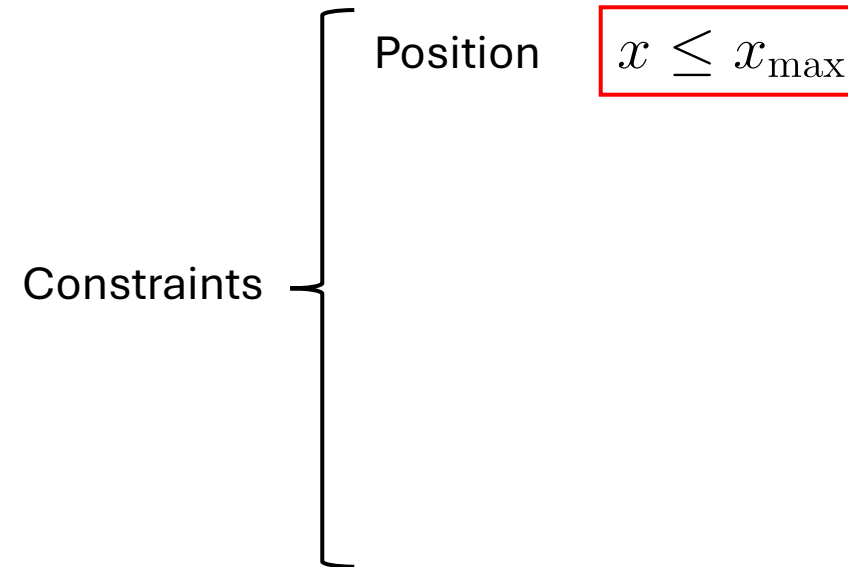
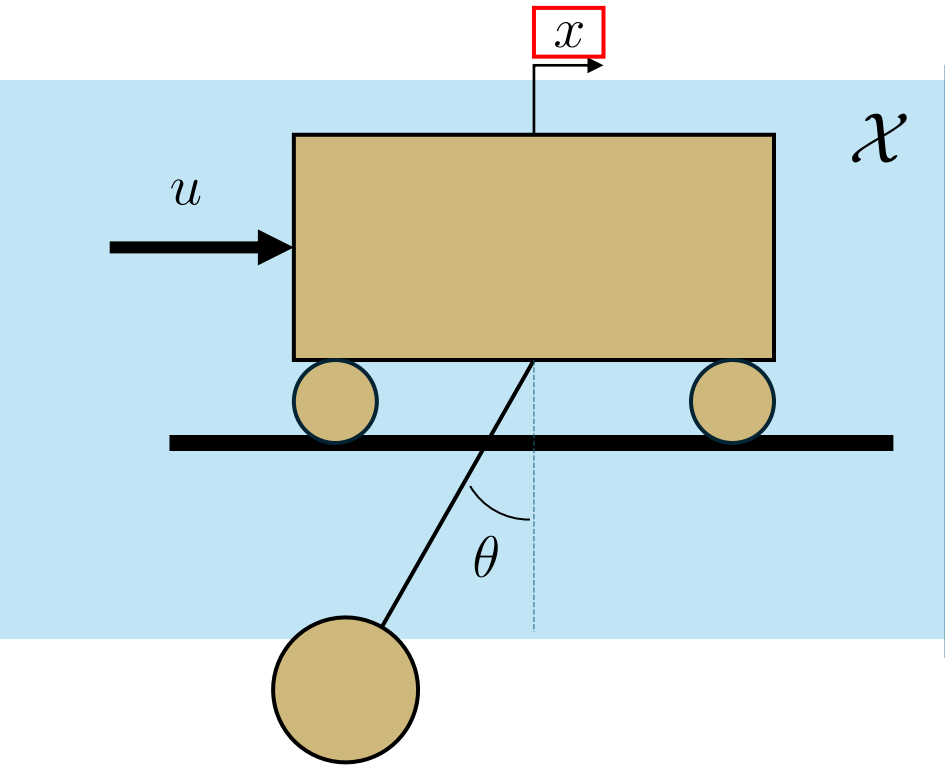
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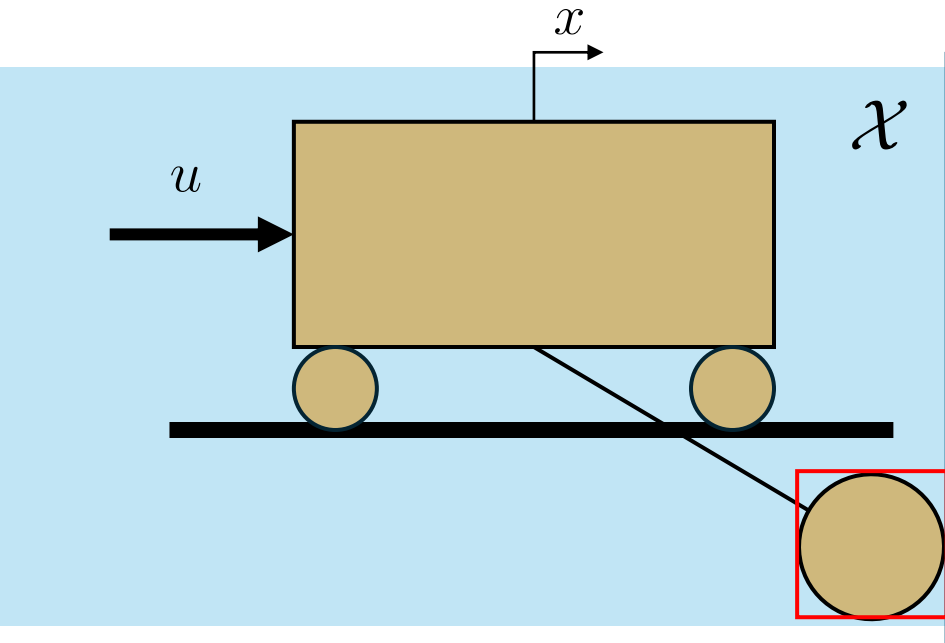
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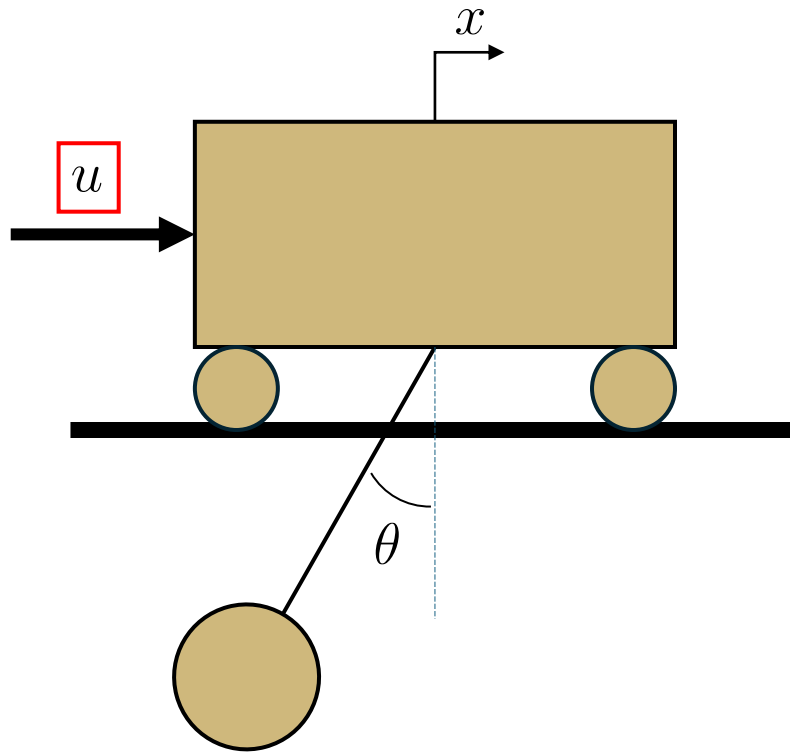
Example: Overhead Crane



Constraints

- Position $x \leq x_{\max}$
- Payload $x + L \sin \theta \leq p_{\max}$

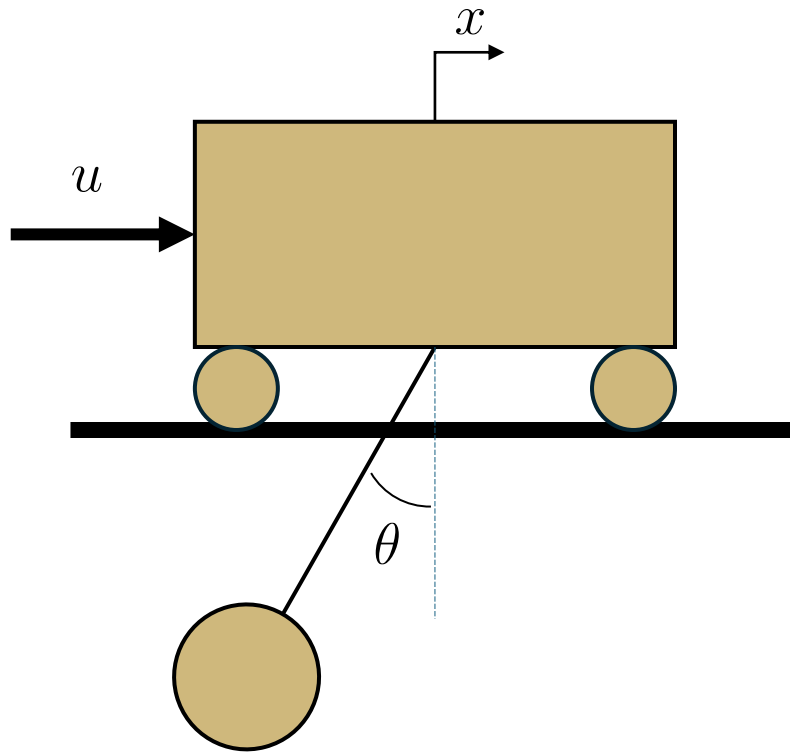
Example: Overhead Crane



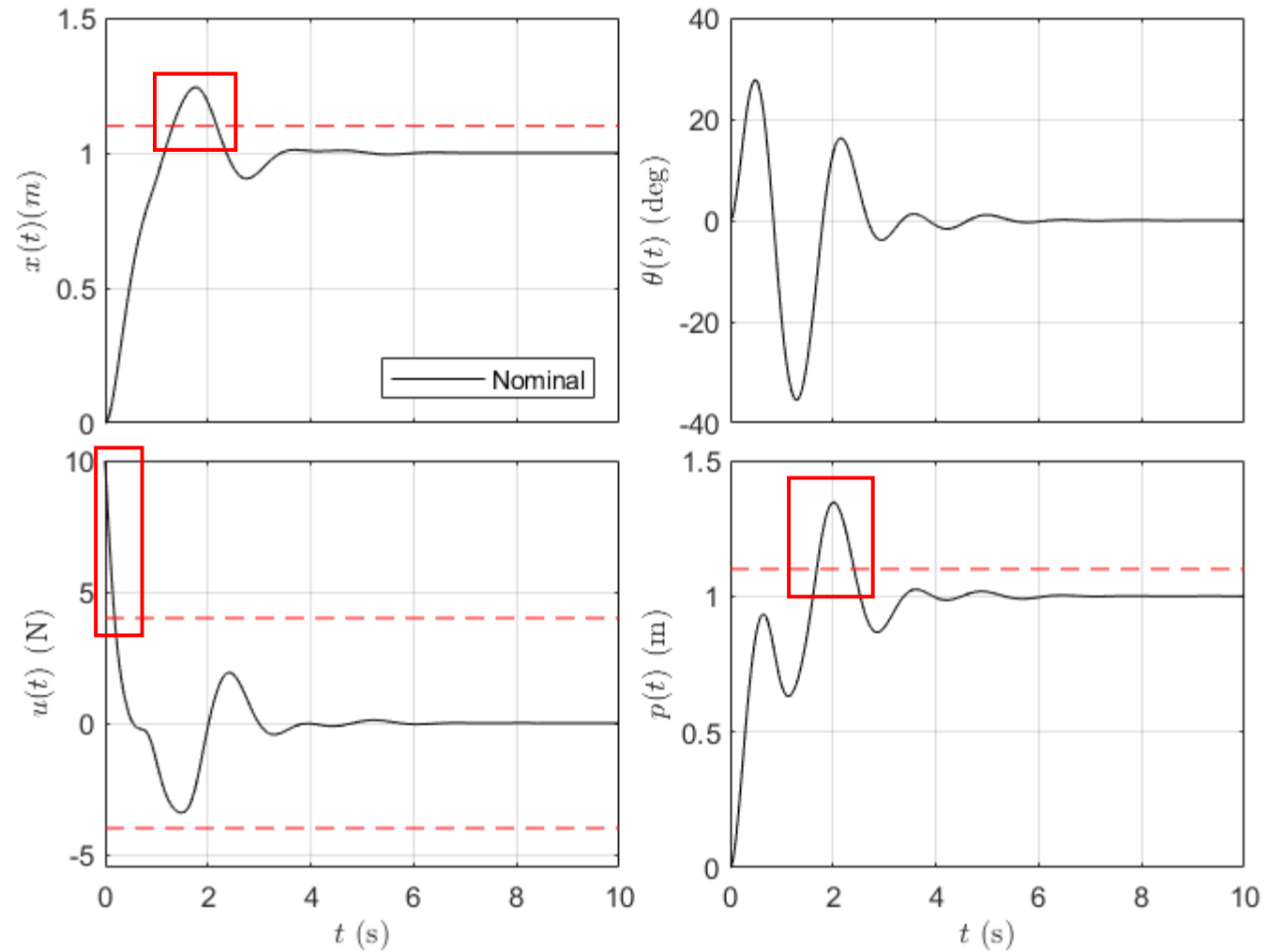
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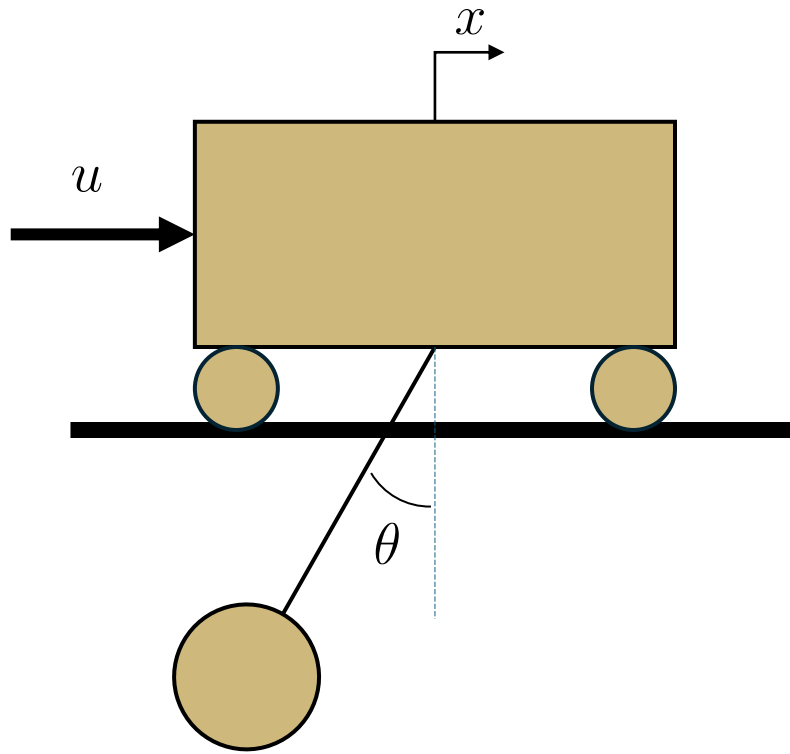
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Example: Overhead Crane

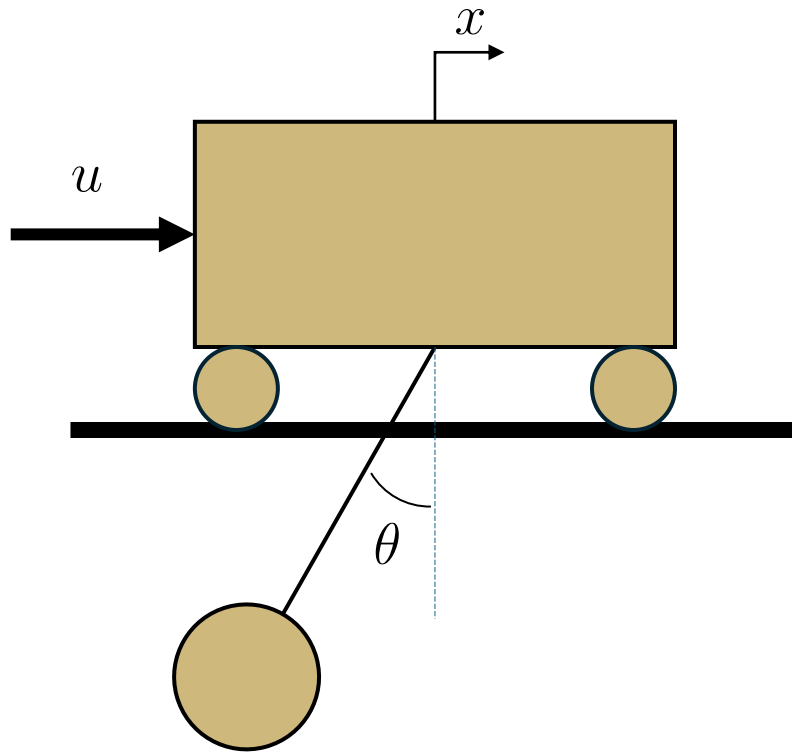


Design candidate CBFs

Constraints

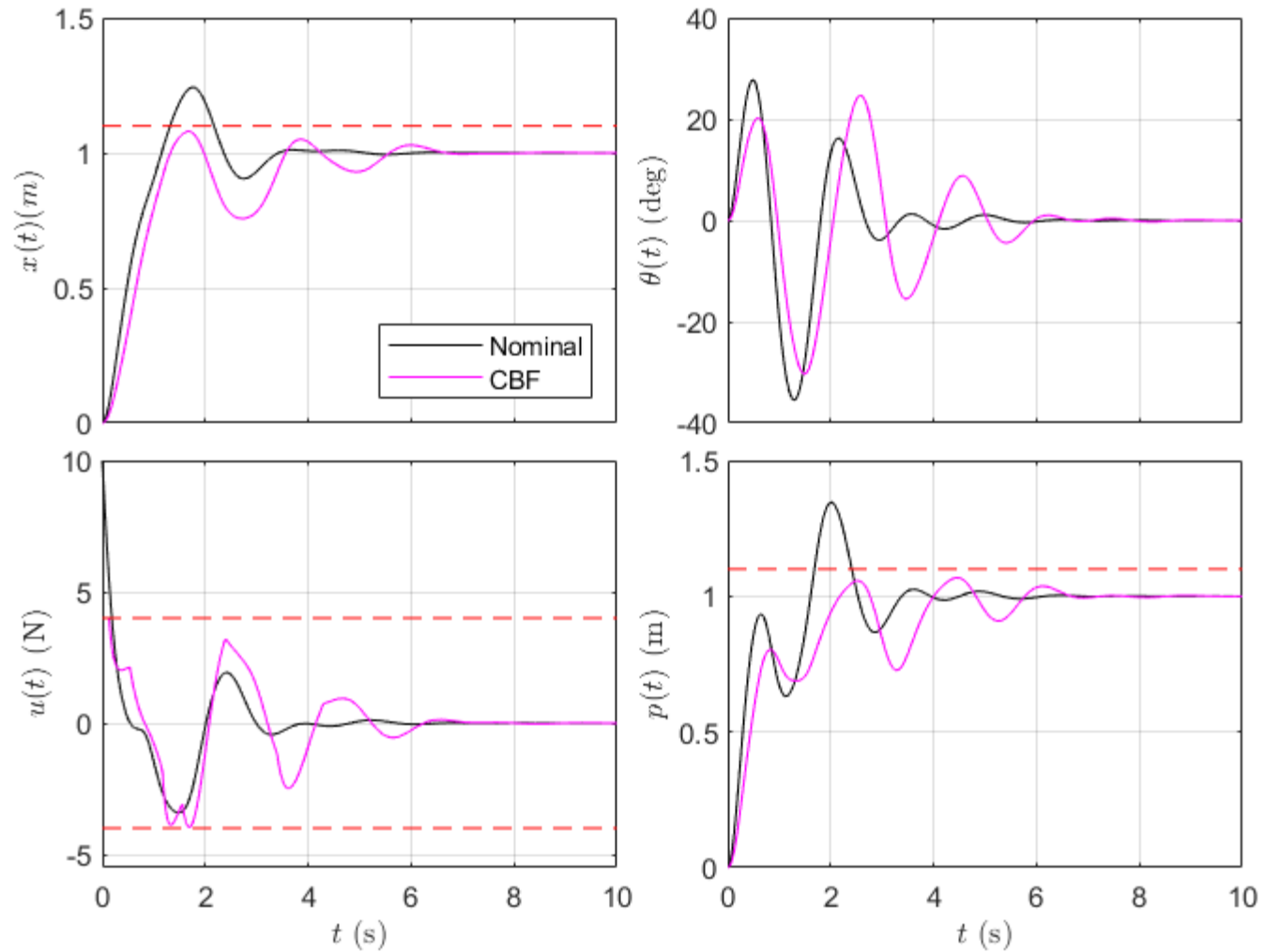
- Position $x \leq x_{\max}$
 $h_1(x) = x_{\max} - x$
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Example: Overhead Crane

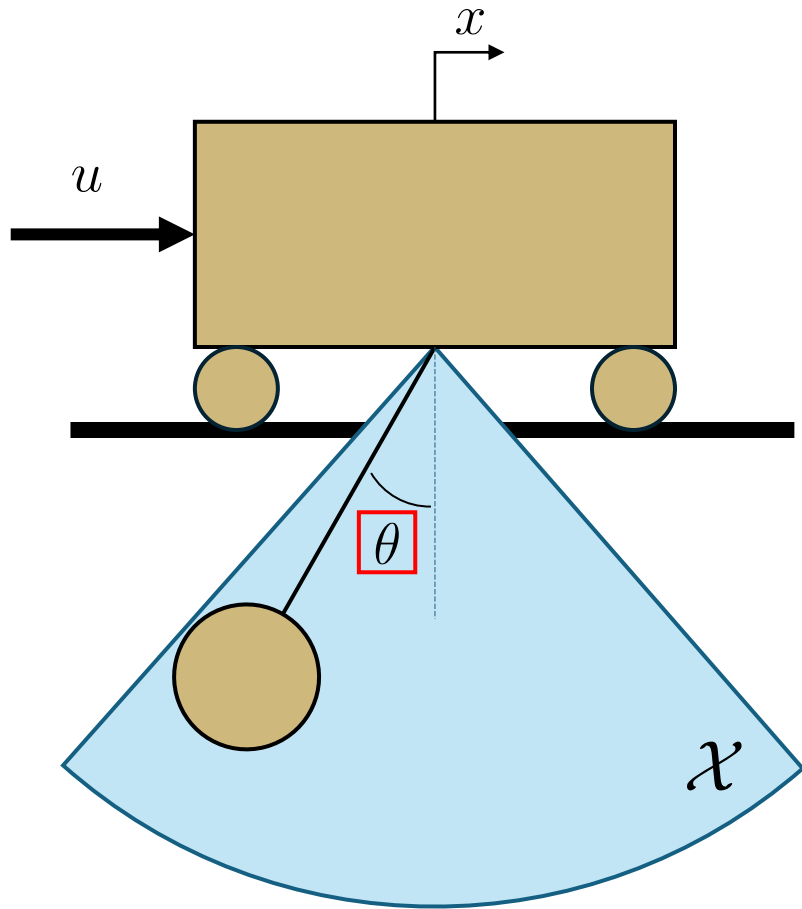


CBF-based control

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 \\ \text{s.t.} \quad & \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x})) \end{aligned}$$



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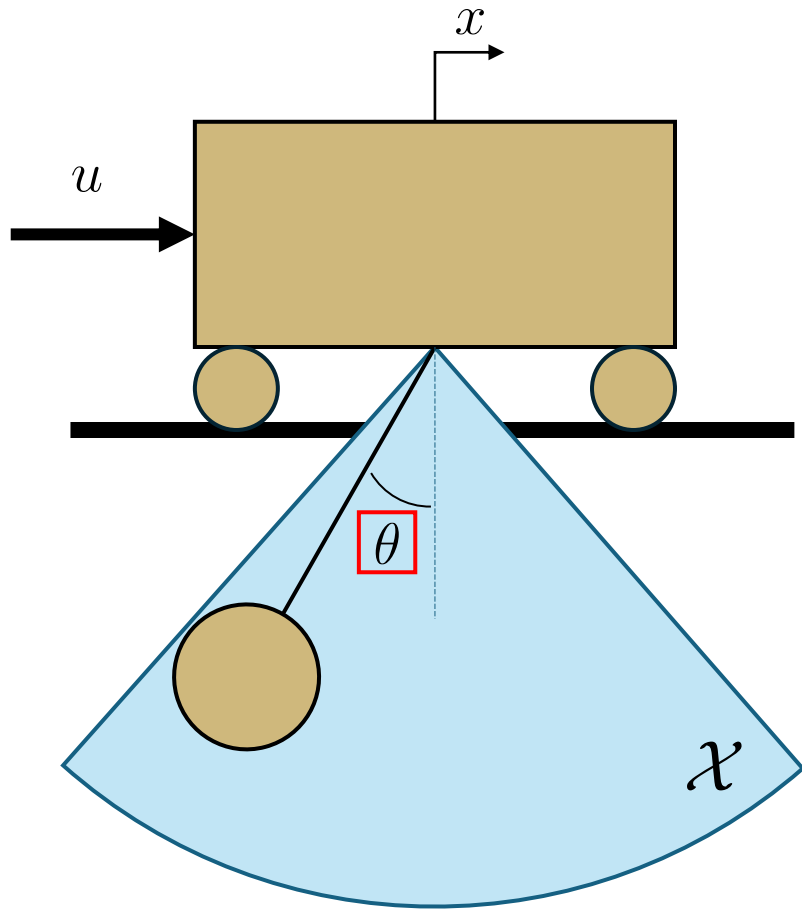


Add new constraint

Constraints

Position	$x \leq x_{\max}$ $h_1(x) = x_{\max} - x$
Payload	$x + L \sin \theta \leq p_{\max}$ $h_2(x, \theta) = p_{\max} - x - L \sin \theta$
Input	$ u \leq u_{\max}$
Angle	$ \theta \leq \theta_{\max}$

Example: Overhead Crane

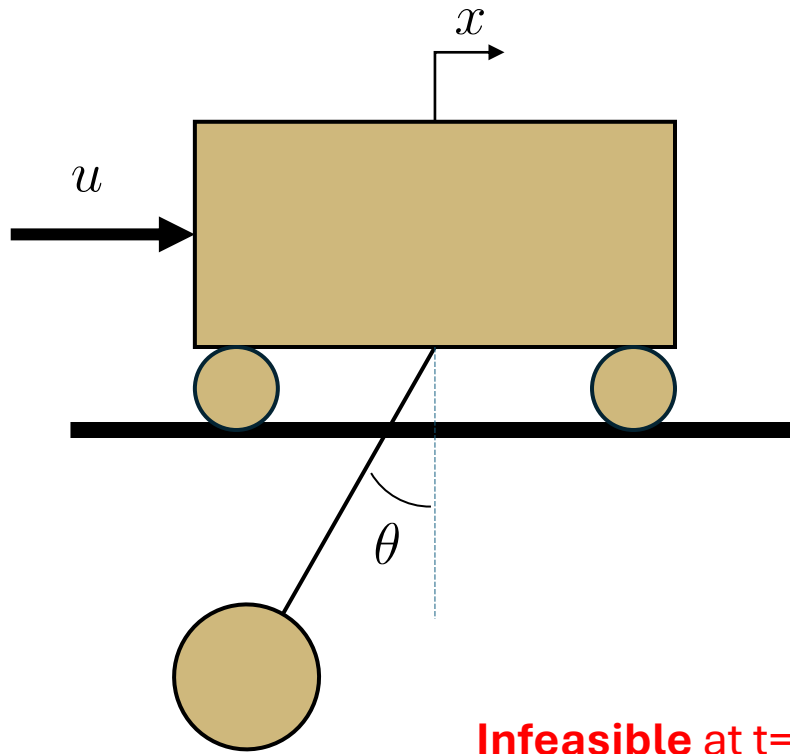


Design candidate CBFs

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- Angle $|\theta| \leq \theta_{\max}$
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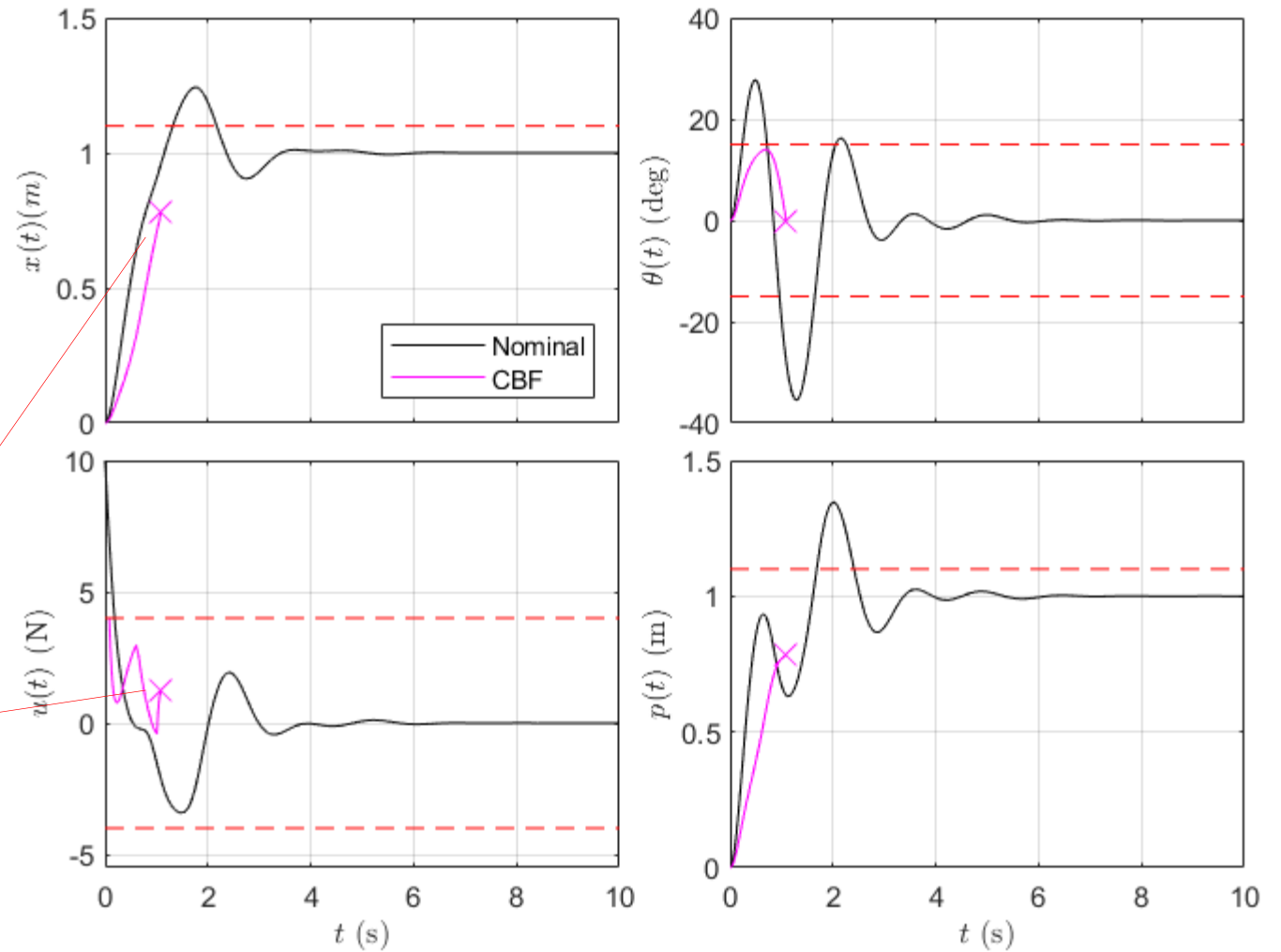
Example: Overhead Crane



CBF-based control

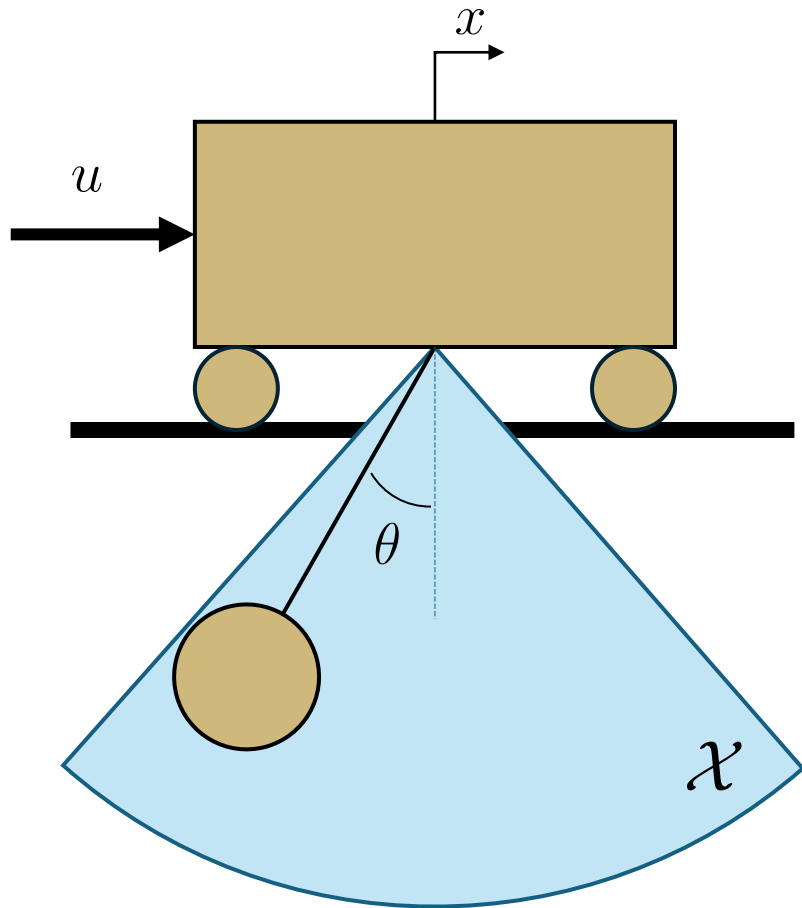
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Infeasible at $t=1.1$



Example: Overhead Crane

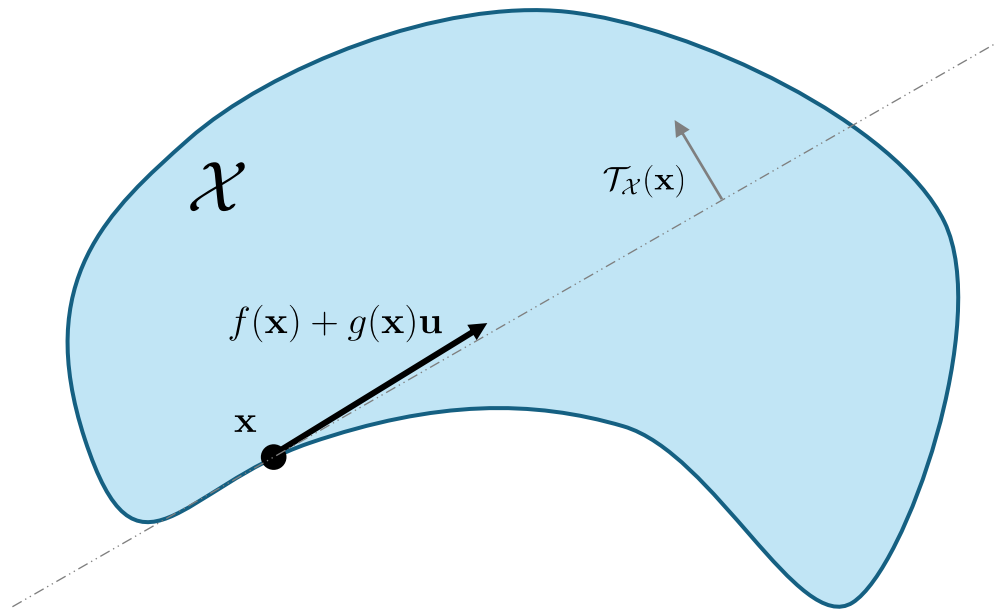
Limitation of naïve (candidate) CBF design: **CBF-QP Infeasible**



Constraints

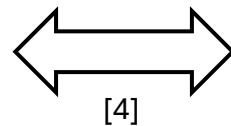
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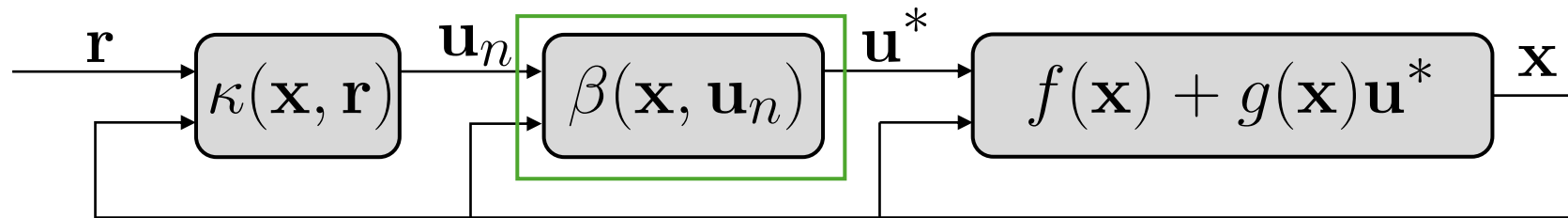
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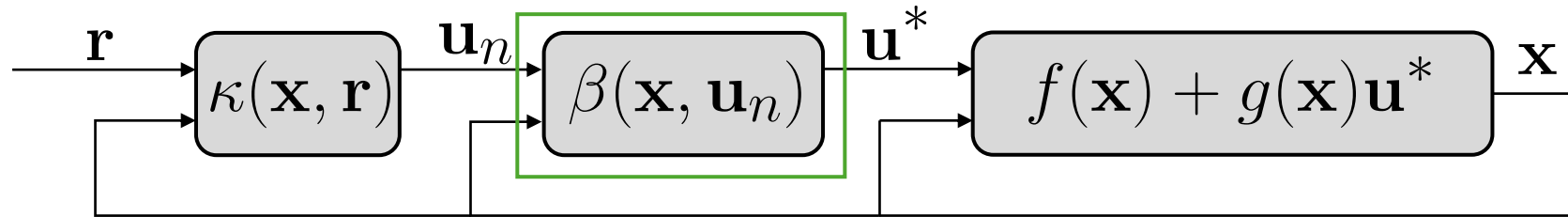
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Reference Governor (RG) Approaches

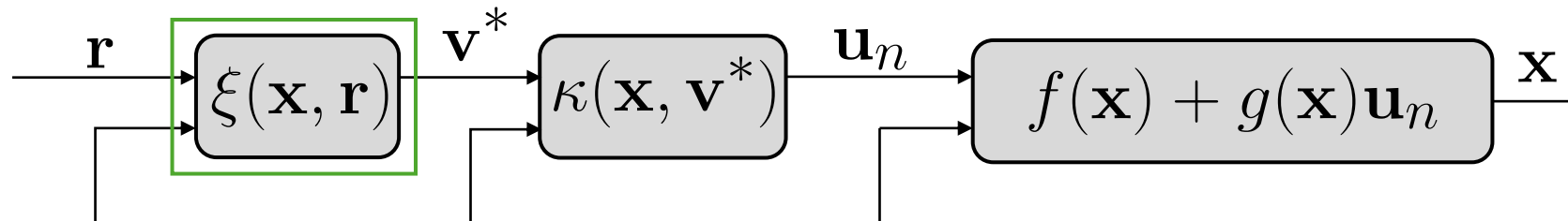


CBF-based control

Reference Governor (RG) Approaches

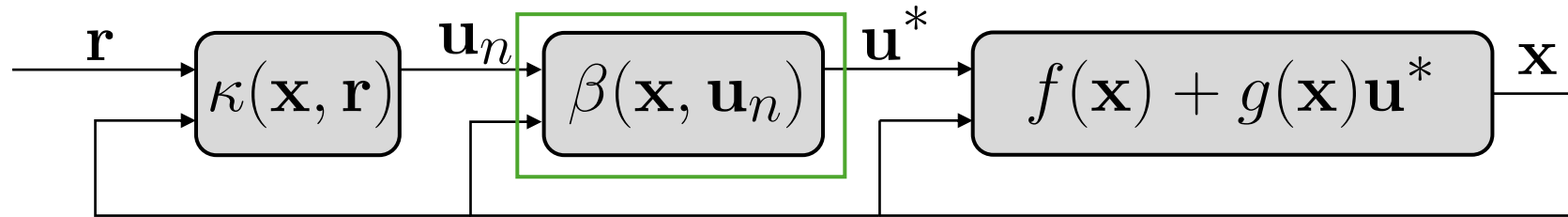


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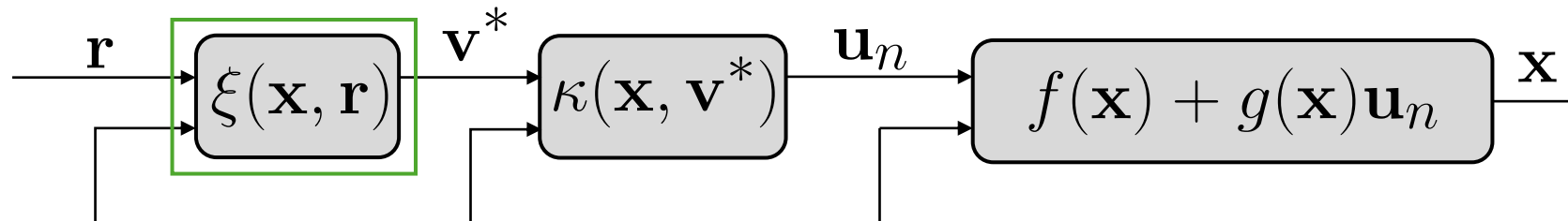
Reference Governor (RG) Approaches



CBF-based control

Good performance

No systematic construction



RG-based control

Poor performance

Systematic construction

Dynamic Safety Margin (DSM)

Continuous-time nonlinear system:

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Prestabilizing controller:

$$\mathbf{u} = \kappa(\mathbf{x}, \mathbf{v})$$



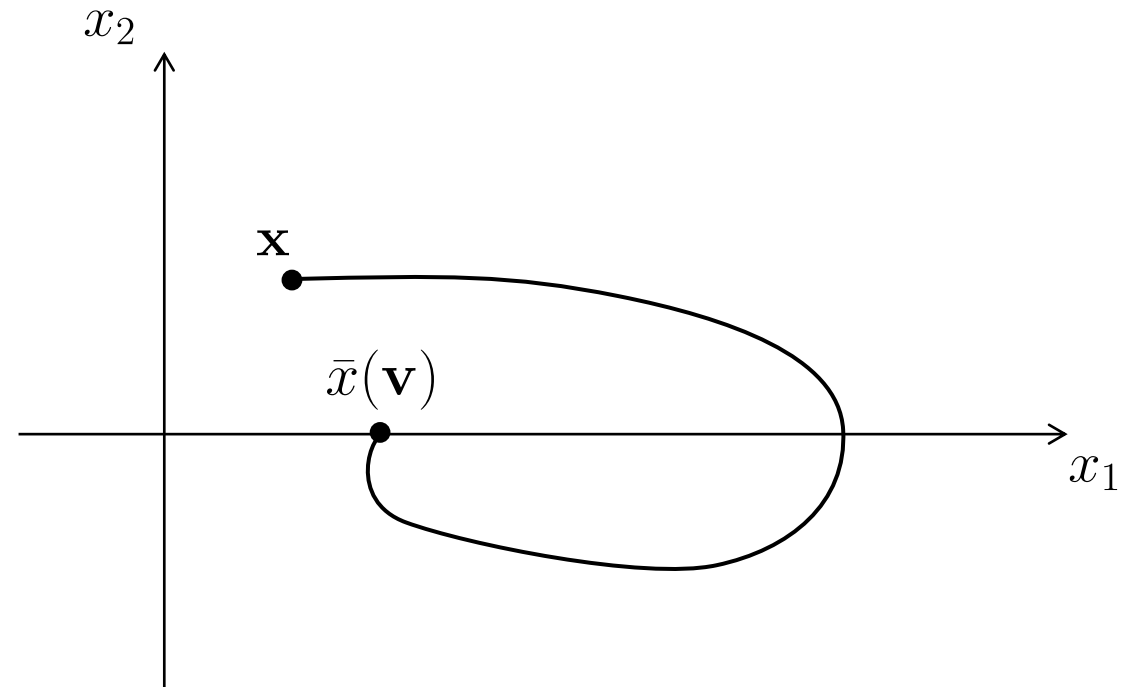
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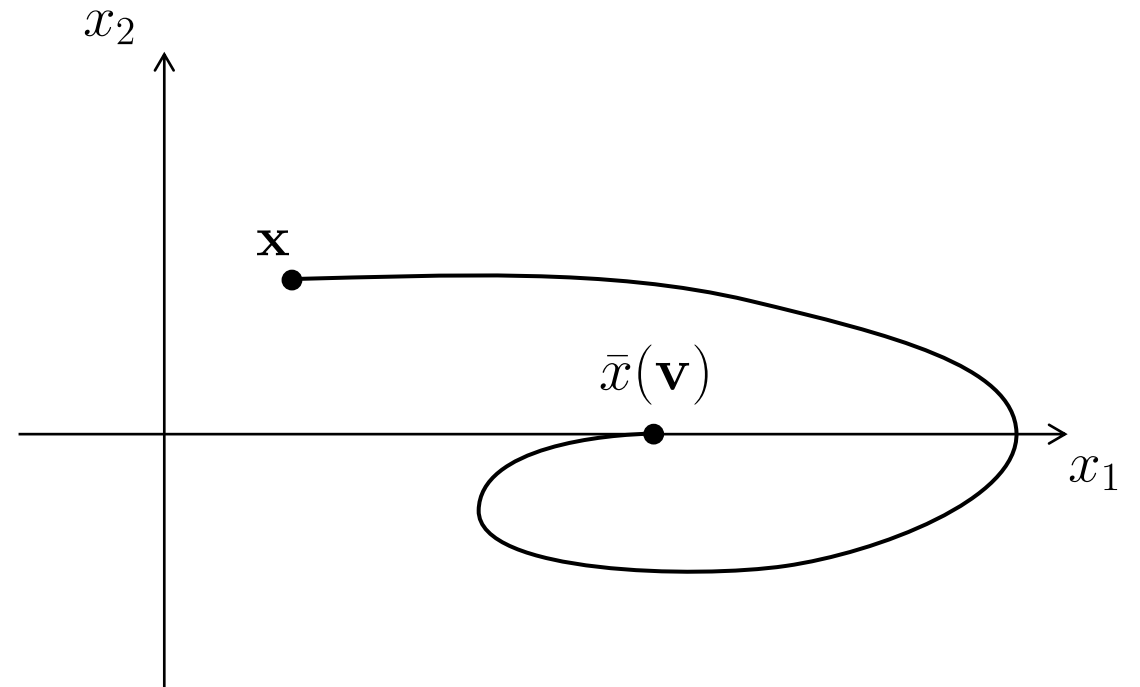
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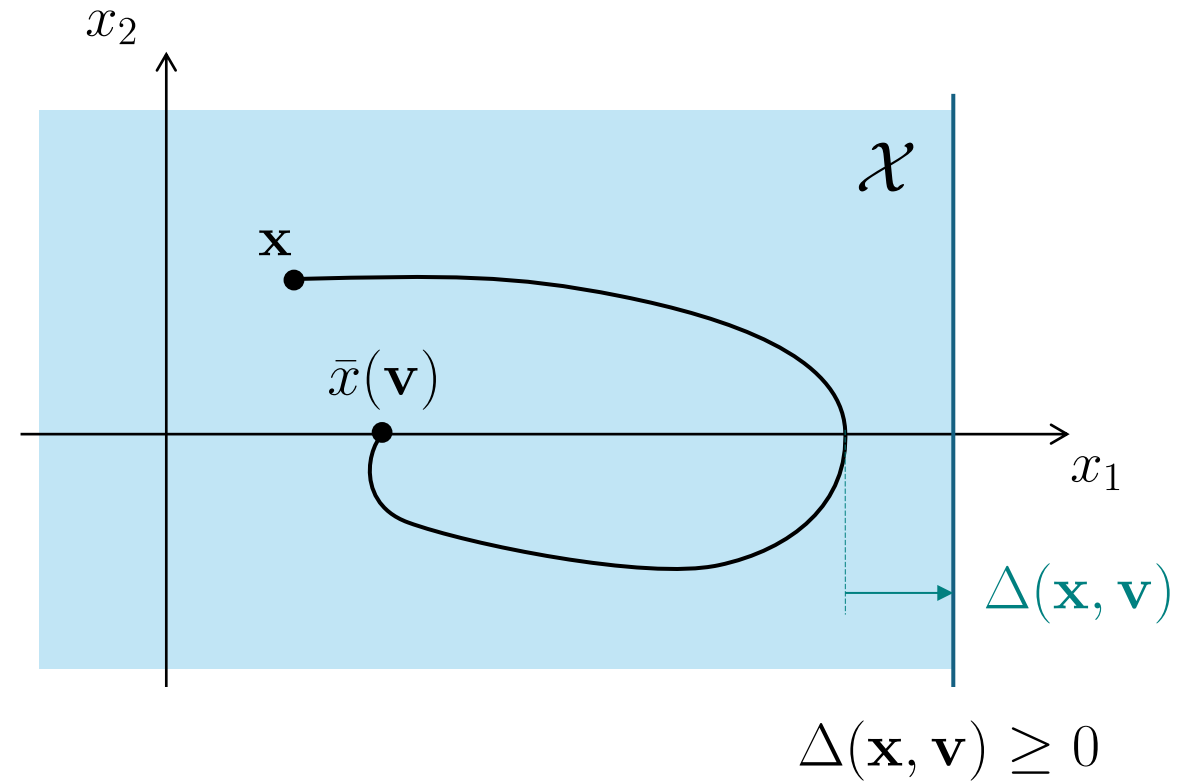
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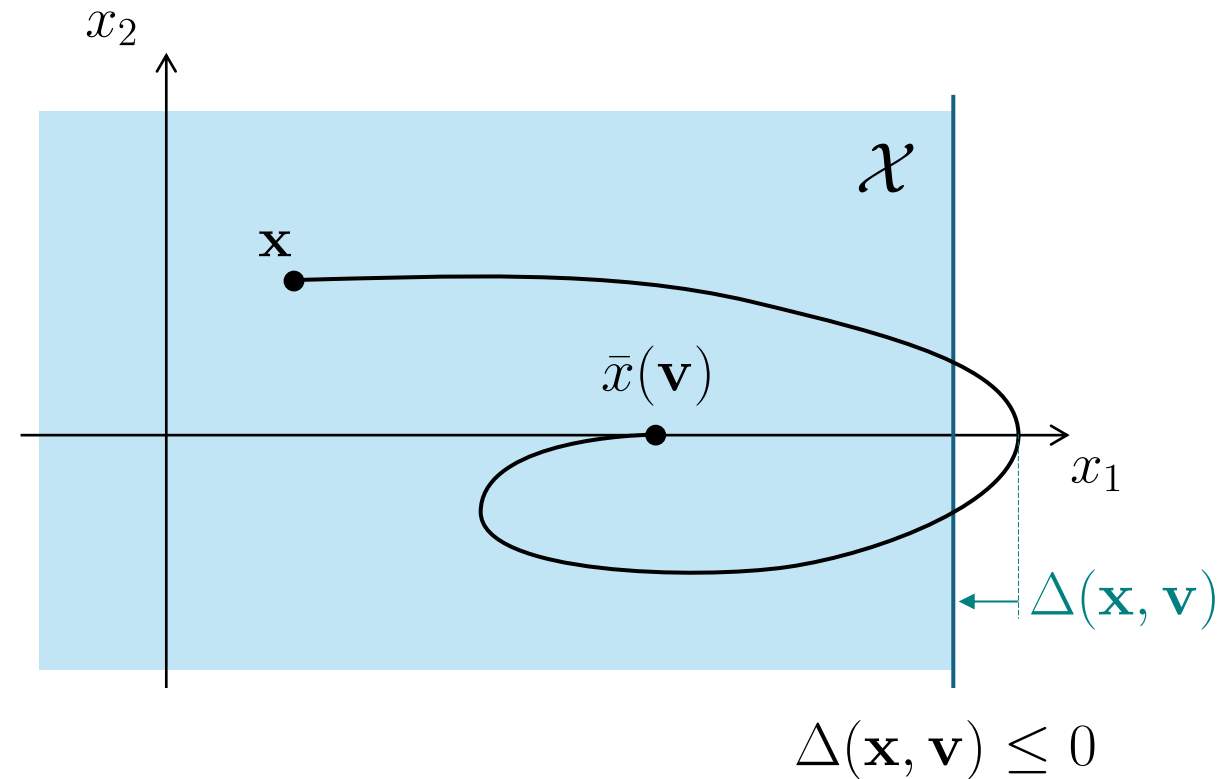
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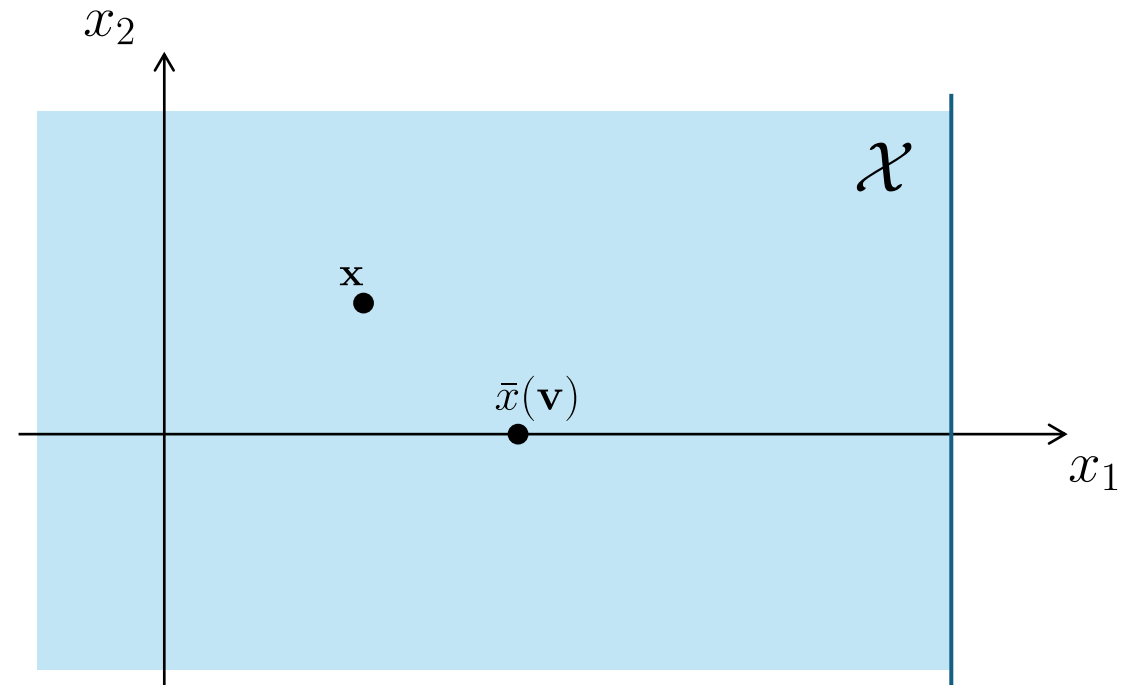
Lyapunov-based DSMs

Reference-dependent Lyapunov function

$$V(\bar{x}(\mathbf{v}), \mathbf{v}) = 0$$

$$V(\mathbf{x}, \mathbf{v}) > 0$$

$$\frac{\partial V}{\partial \mathbf{x}}(f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v})) \leq 0$$



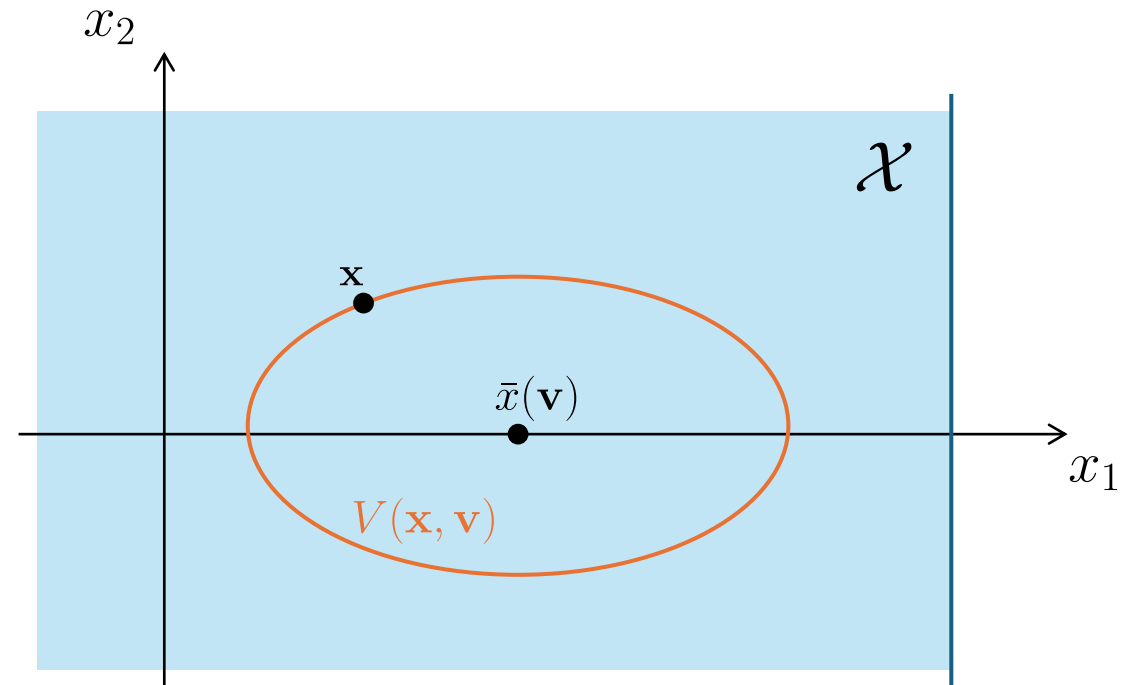
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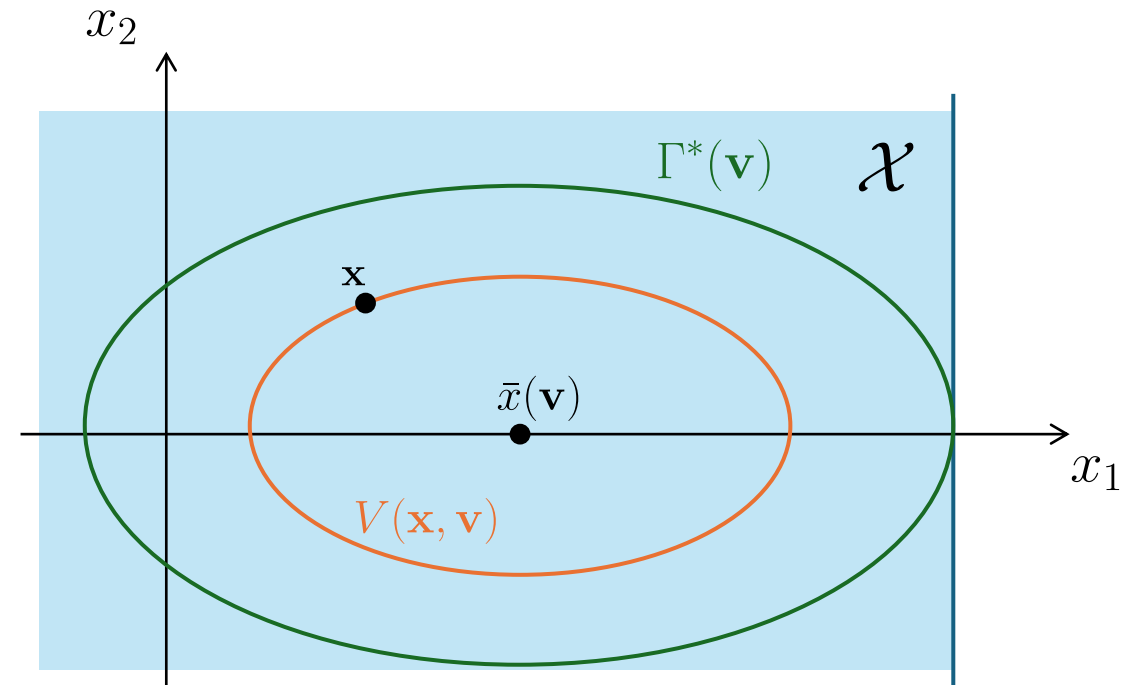
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Threshold value $\Gamma^*(\mathbf{v})$



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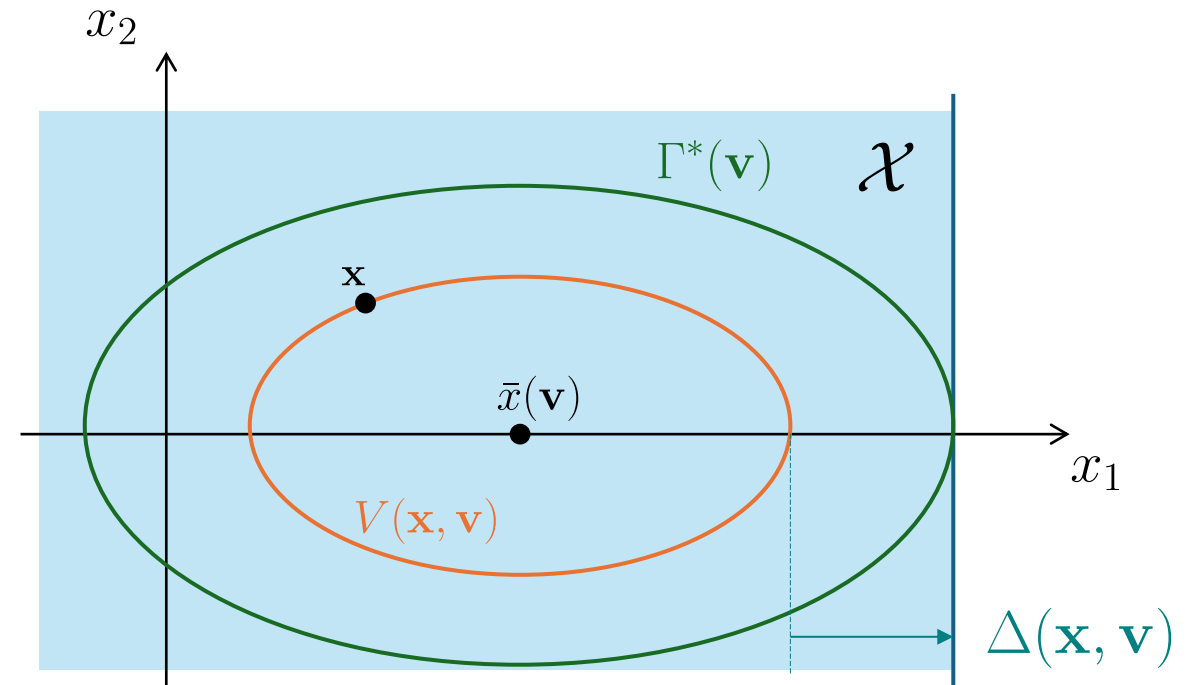
$$\frac{\partial V}{\partial \mathbf{x}}(f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v})) \leq 0$$

Threshold value $\Gamma^*(\mathbf{v})$

Lyapunov-based DSM:

$$\Delta(\mathbf{x}, \mathbf{v}) = \Gamma^*(\mathbf{v}) - V(\mathbf{x}, \mathbf{v})$$

Other approaches to construct DSMs in [5]



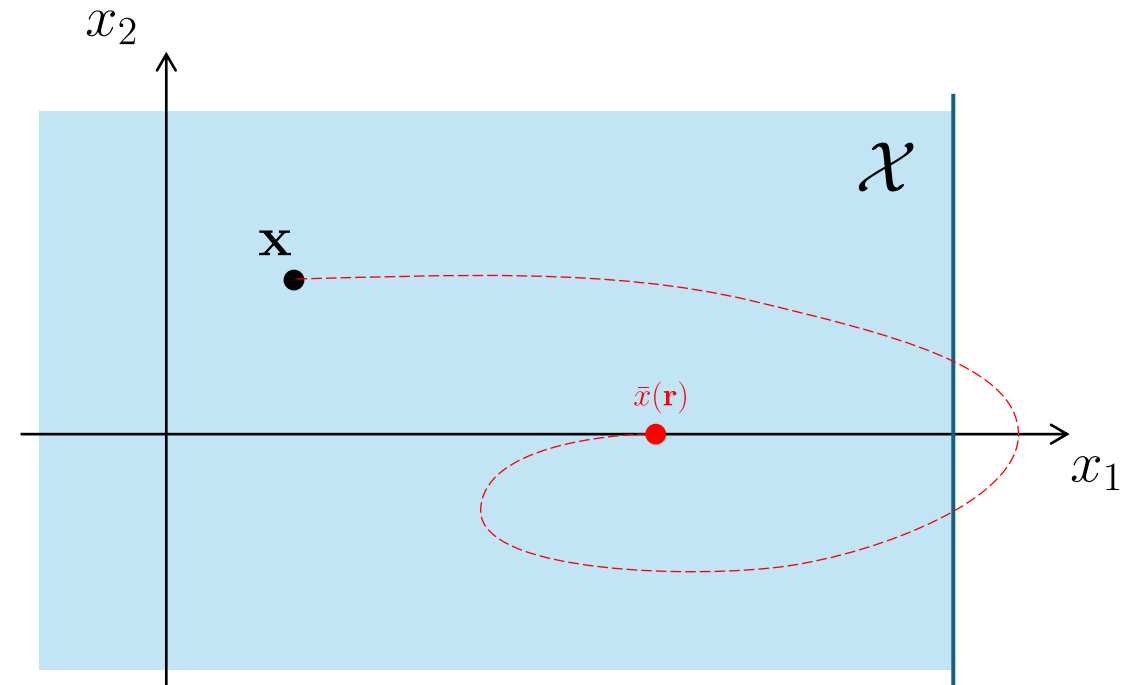
Explicit Reference Governor (ERG)

Want to reach $\bar{x}(\mathbf{r})$, but it is unsafe with $\mathbf{u} = \kappa(\mathbf{x}, \mathbf{r})$

ERG closed-loop system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$

Guarantees $\mathbf{x}(t) \in \mathcal{X}$



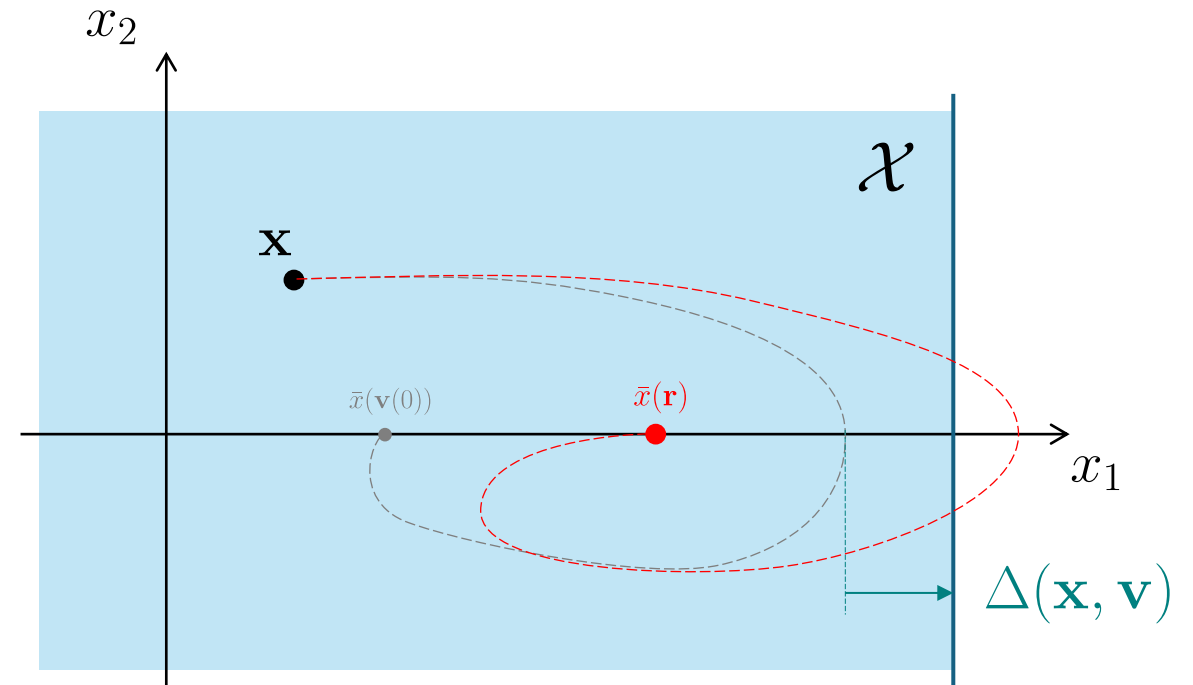
Explicit Reference Governor (ERG)

Initialize $\mathbf{v}$ so that $\Delta(\mathbf{x}, \mathbf{v}) \geq 0$

ERG closed-loop system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$

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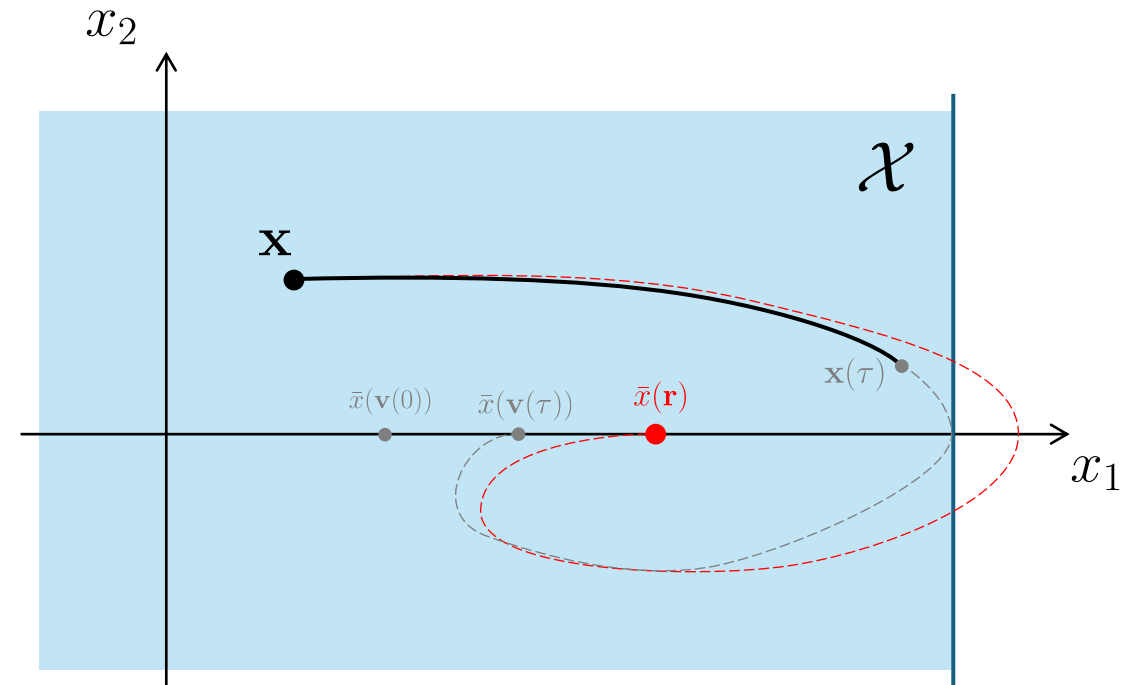
Explicit Reference Governor (ERG)

System evolves and $\mathbf{v} \rightarrow \mathbf{r}$ as long as $\Delta(\mathbf{x}, \mathbf{v}) > 0$

ERG closed-loop system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$

Guarantees $\mathbf{x}(t) \in \mathcal{X}$



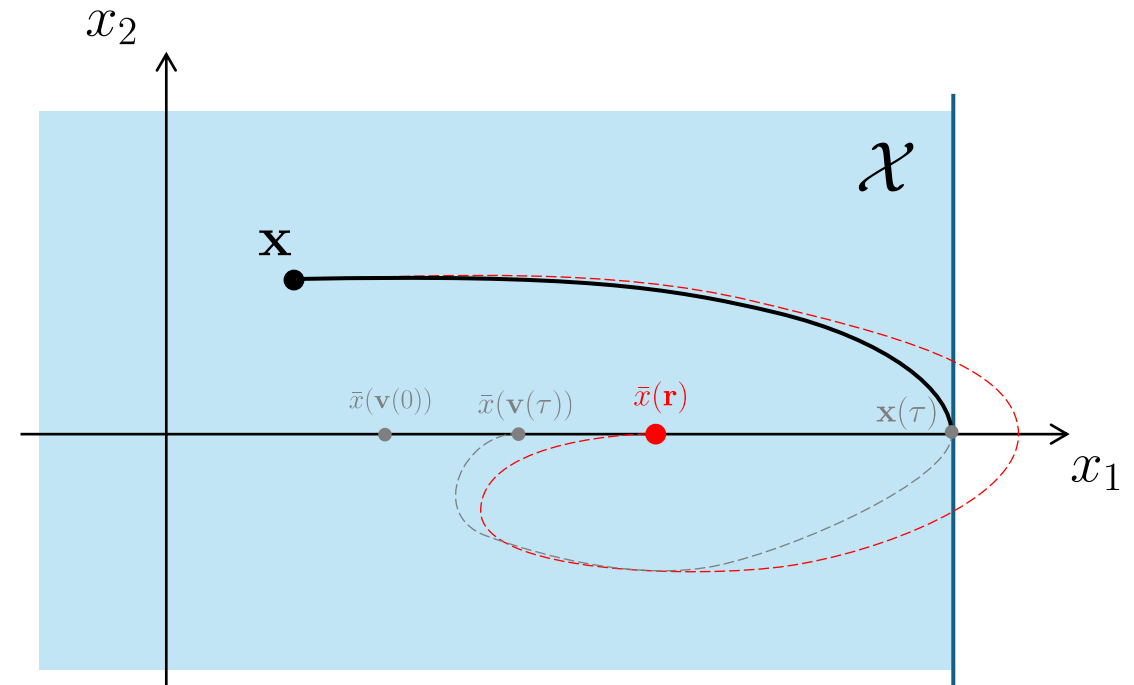
Explicit Reference Governor (ERG)

$\mathbf{v}$ remains constant when $\Delta(\mathbf{x}, \mathbf{v}) = 0$

ERG closed-loop system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$

Guarantees $\mathbf{x}(t) \in \mathcal{X}$



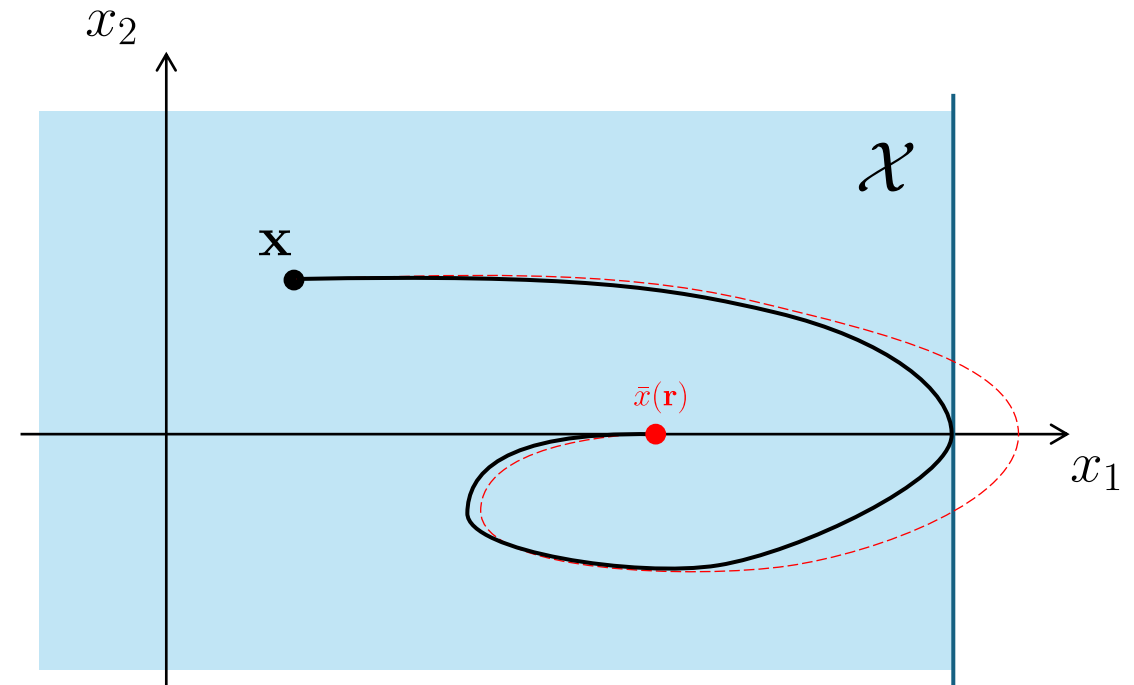
Explicit Reference Governor (ERG)

Eventually, $\mathbf{x}(t) \rightarrow \bar{\mathbf{x}}(\mathbf{r})$ and $\mathbf{v} \rightarrow \mathbf{r}$

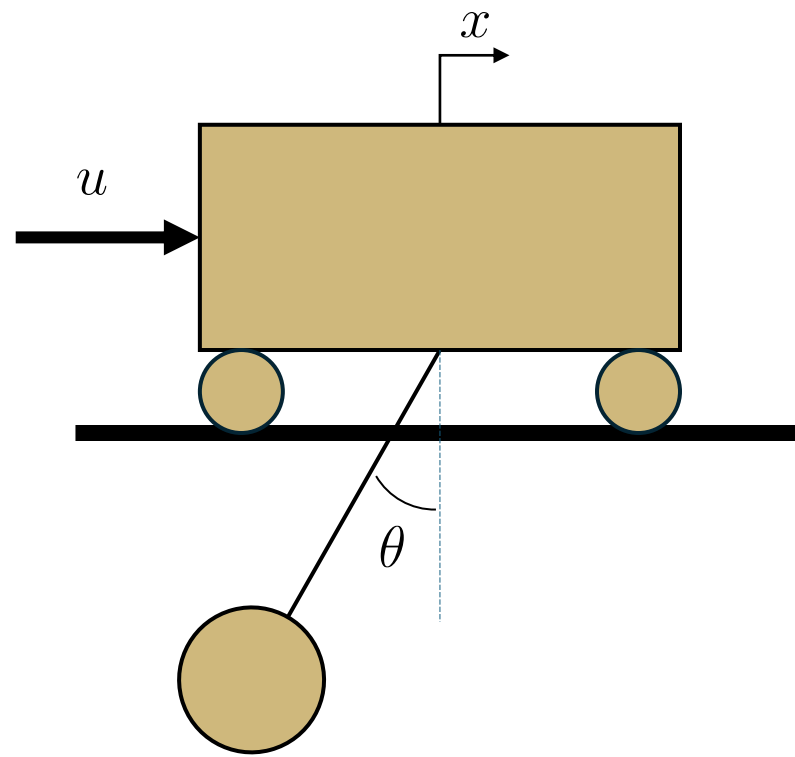
ERG closed-loop system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$

Guarantees $\mathbf{x}(t) \in \mathcal{X}$

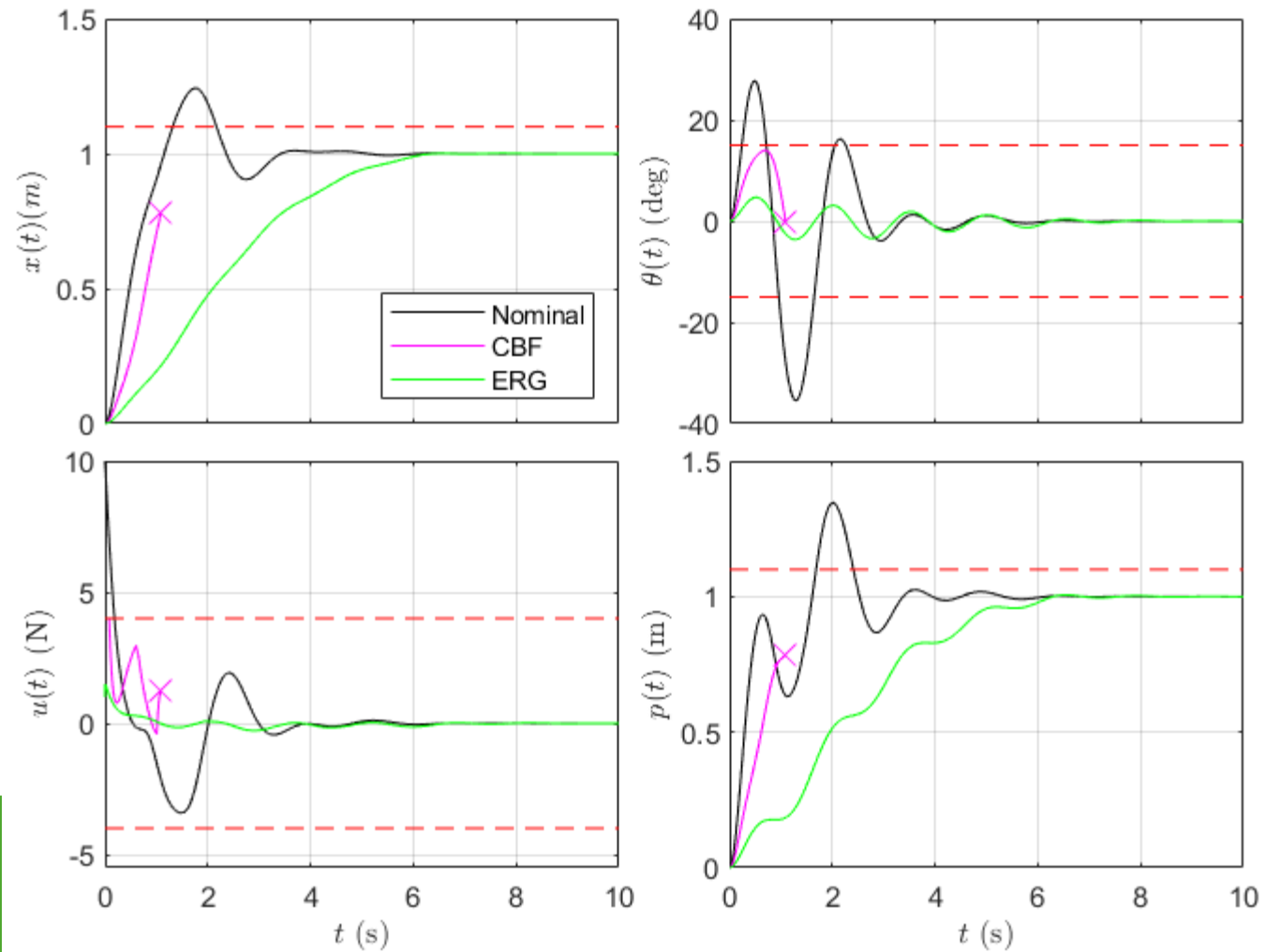


Example: Overhead Crane



ERG [5]

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\kappa(\mathbf{x}, \mathbf{v}) \\ \Delta(\mathbf{x}, \mathbf{v})(\mathbf{r} - \mathbf{v}) \end{bmatrix}$$



Our work: DSM-CBF



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Theorem: DSMs are CBFs for the augmented system

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \\ \mathbf{w} \end{bmatrix}$$

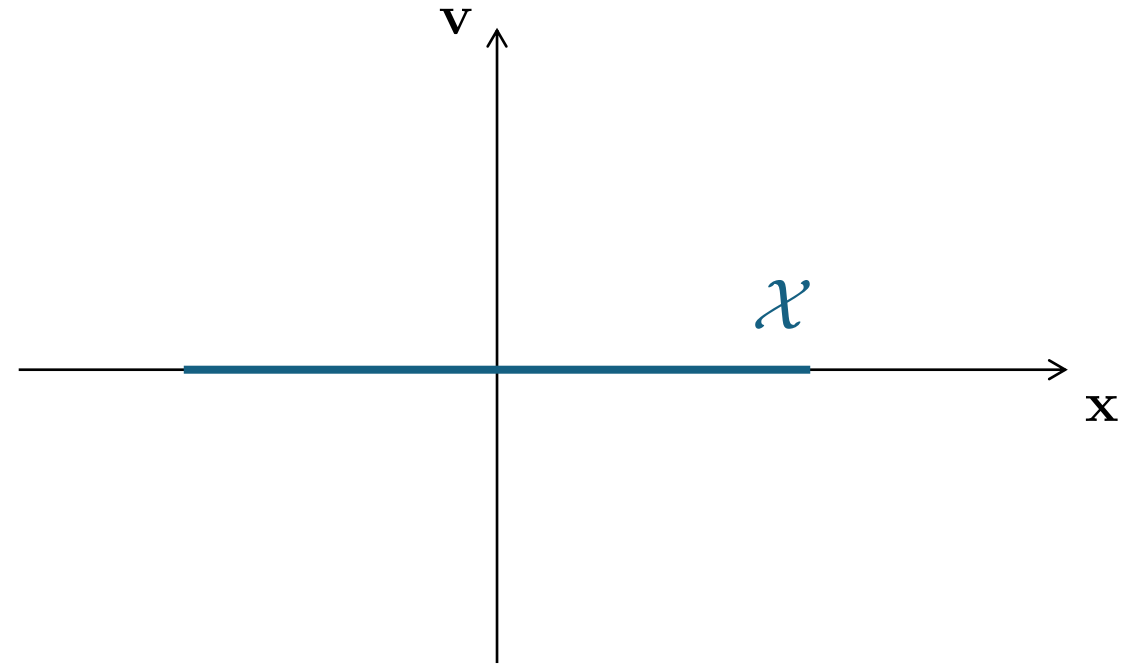


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Intuition:

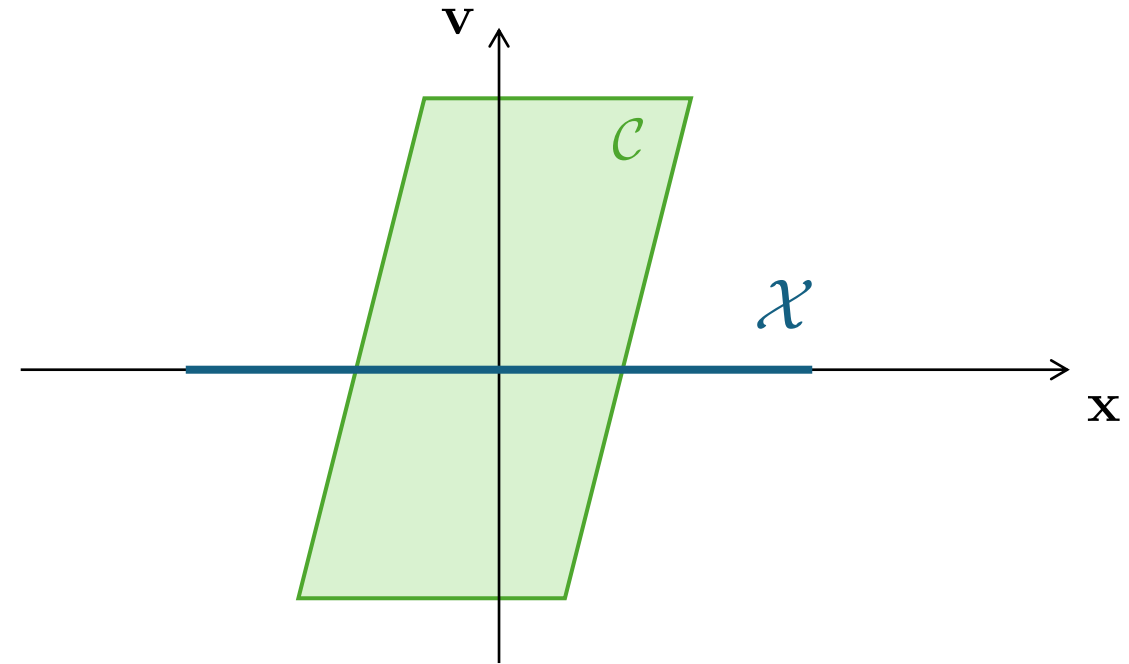


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Intuition:



$$\mathcal{C} = \{(\mathbf{x}, \mathbf{v}) \mid \Delta(\mathbf{x}, \mathbf{v}) \geq 0\}$$

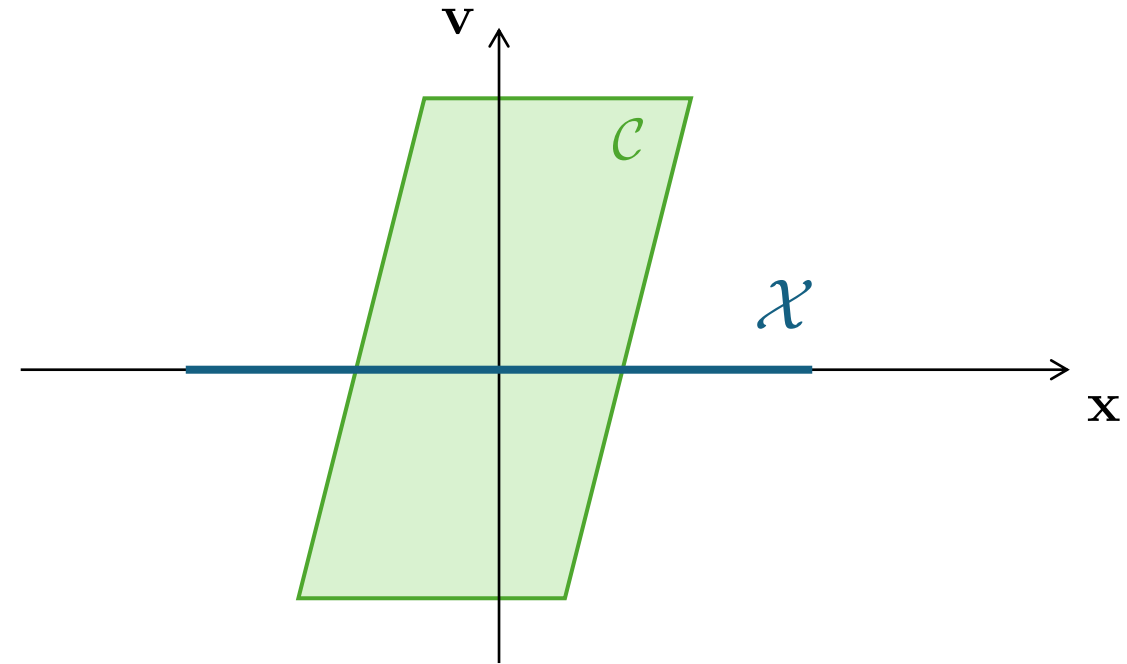
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$\mathcal{C}$ is control invariant because the input
 $(\mathbf{u}, \mathbf{w}) = (\kappa(\mathbf{x}, \mathbf{v}), 0)$
is always safe (satisfies CBF-condition)

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Our work: DSM-CBF

Theorem: DSMs are CBFs for the augmented system

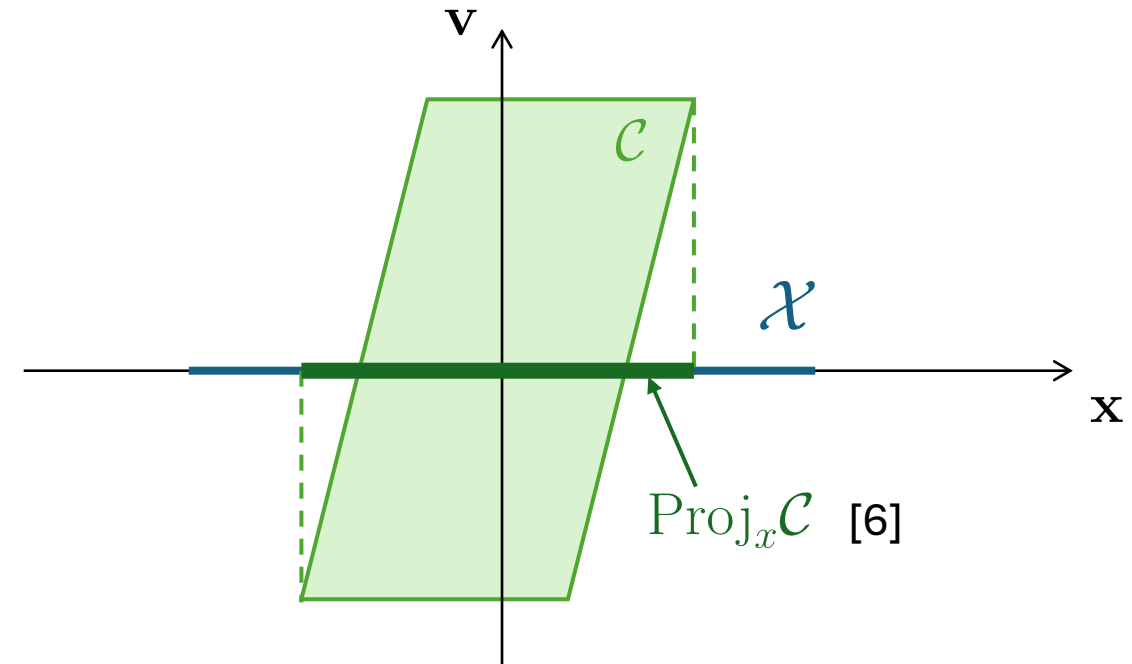
$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{v}} \end{bmatrix} = \begin{bmatrix} f(\mathbf{x}) + g(\mathbf{x})\mathbf{u} \\ \mathbf{w} \end{bmatrix}$$

$\mathcal{C}$ is control invariant because the input
 $(\mathbf{u}, \mathbf{w}) = (\kappa(\mathbf{x}, \mathbf{v}), 0)$
is always safe (satisfies CBF-condition)

$\text{Proj}_x \mathcal{C}$ is control invariant with respect to the original system

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}$$

Intuition:



$$\mathcal{C} = \{(\mathbf{x}, \mathbf{v}) \mid \Delta(\mathbf{x}, \mathbf{v}) \geq 0\}$$

Our work: DSM-CBF-based Control

DSM-CBF control

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 + \eta \|\mathbf{w} - \mathbf{w}_n\|^2 \\ \text{s.t.} \quad & \dot{\Delta}(\mathbf{x}, \mathbf{v}, \mathbf{u}, \mathbf{w}) \geq -\alpha(\Delta(\mathbf{x}, \mathbf{v})) \end{aligned}$$



Our work: DSM-CBF-based Control

DSM-CBF control

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CBF control

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 \\ \text{s.t.} \quad & \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x})) \end{aligned}$$



Our work: DSM-CBF-based Control

DSM-CBF control

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 + \eta \|\mathbf{w} - \mathbf{w}_n\|^2 \\ \text{s.t.} \quad & \dot{\Delta}(\mathbf{x}, \mathbf{v}, \mathbf{u}, \mathbf{w}) \geq -\alpha(\Delta(\mathbf{x}, \mathbf{v})) \end{aligned}$$

$$\dot{\Delta}(\mathbf{x}, \mathbf{v}, \mathbf{u}, \mathbf{w}) = \frac{\partial \Delta}{\partial \mathbf{x}} f(\mathbf{x}) + \frac{\partial \Delta}{\partial \mathbf{x}} g(\mathbf{x}) \mathbf{u} + \frac{\partial \Delta}{\partial \mathbf{v}} \mathbf{w}$$

CBF control

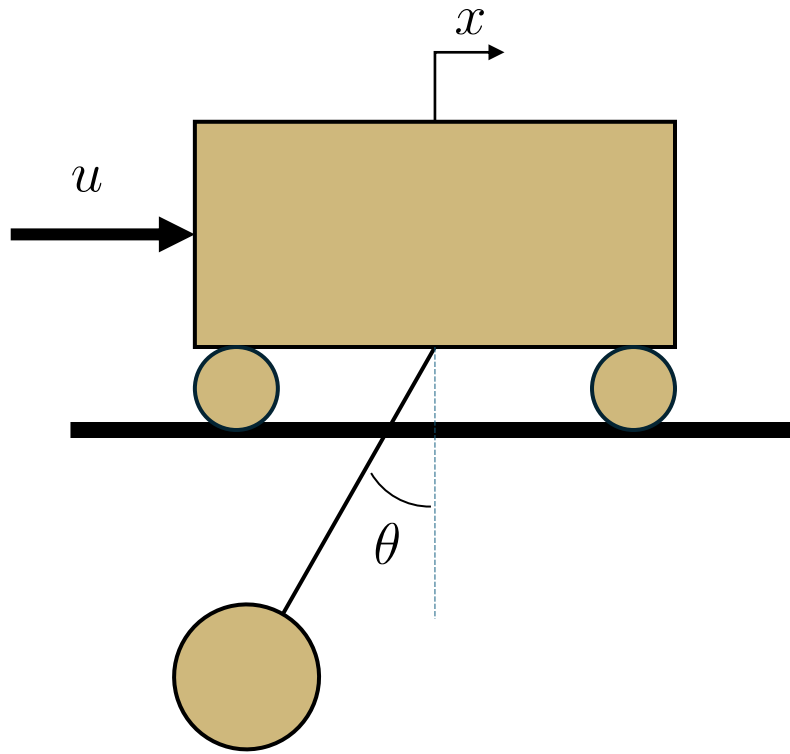
$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 \\ \text{s.t.} \quad & \dot{h}(\mathbf{x}, \mathbf{u}) \geq -\alpha(h(\mathbf{x})) \end{aligned}$$

$$\dot{h}(\mathbf{x}, \mathbf{u}) = \frac{\partial h}{\partial \mathbf{x}} f(\mathbf{x}) + \frac{\partial h}{\partial \mathbf{x}} g(\mathbf{x}) \mathbf{u}$$

Both constraints are linear in decision variables $\rightarrow$ QP

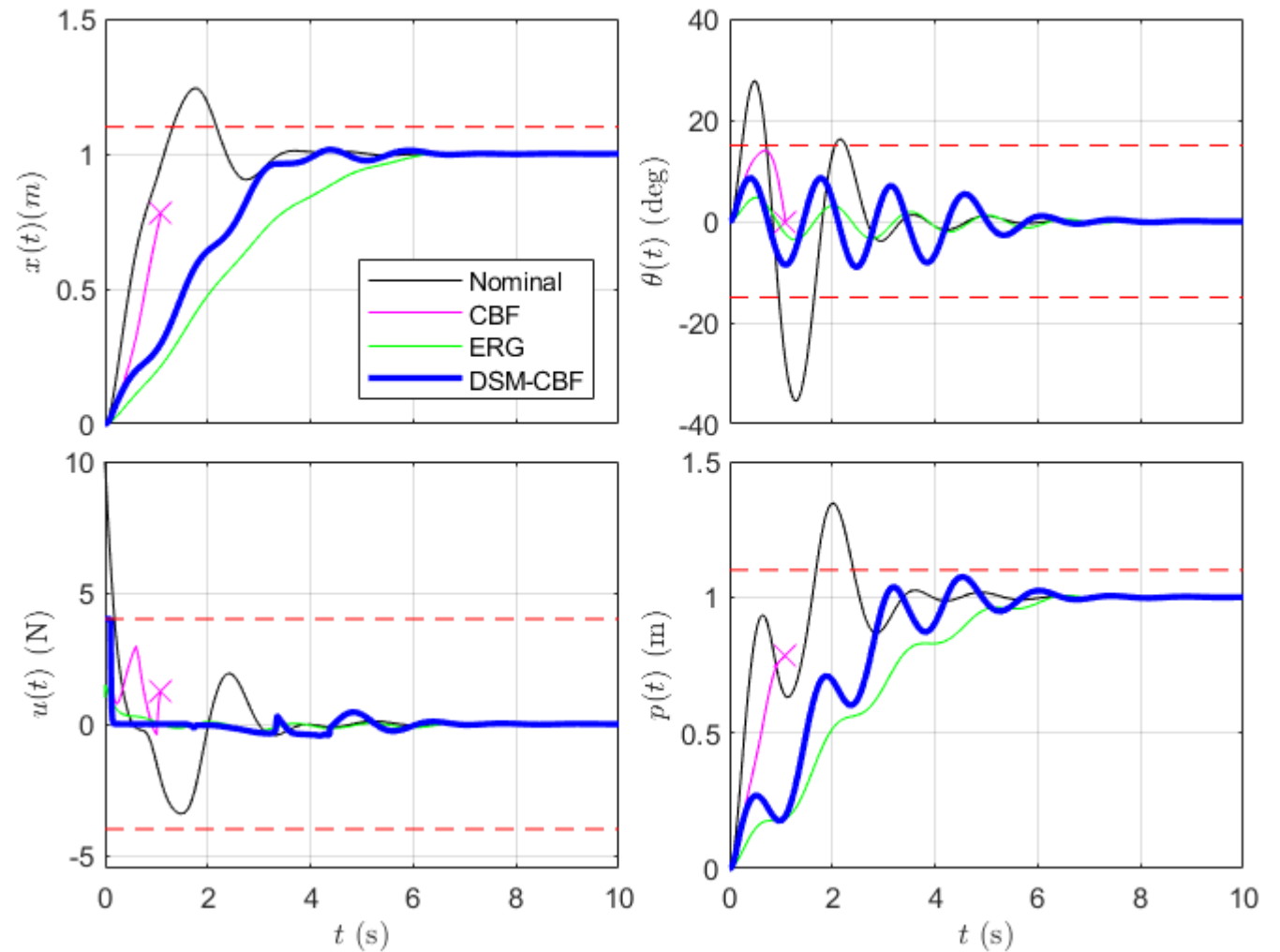


Example: Overhead Crane



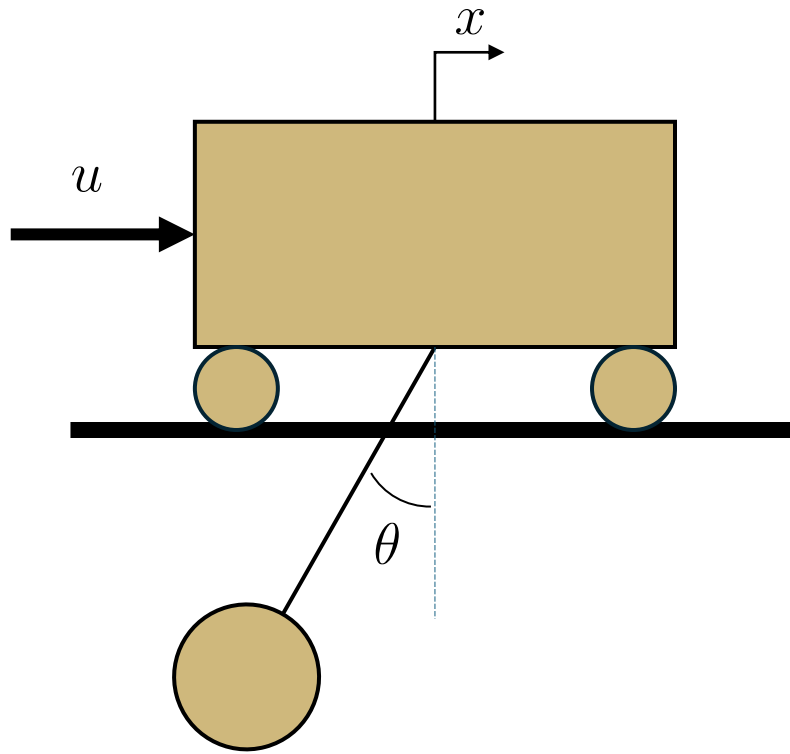
DSM-CBF

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 + \eta \|\mathbf{w} - \mathbf{w}_n\|^2 \\ \text{s.t.} \quad & \dot{\Delta}(\mathbf{x}, \mathbf{v}, \mathbf{u}, \mathbf{w}) \geq -\alpha(\Delta(\mathbf{x}, \mathbf{v})) \end{aligned}$$





Example: Overhead Crane



DSM-CBF

$$\begin{aligned} \min \quad & \|\mathbf{u} - \mathbf{u}_n\|^2 + \eta \|\mathbf{w} - \mathbf{w}_n\|^2 \\ \text{s.t.} \quad & \dot{\Delta}(\mathbf{x}, \mathbf{v}, \mathbf{u}, \mathbf{w}) \geq -\alpha(\Delta(\mathbf{x}, \mathbf{v})) \end{aligned}$$

